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Research paper

Physics-constrained neural networks for reliable solar power forecasting with uncertainty quantification

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ABSTRACT

While accurate forecasting with uncertainty estimation is highly important, Machine Learning (ML) and Deep Learning (DL) models often ignore fundamental physical principles and typically do not provide reliable uncertainty estimates. This study proposes a constraint-guided physics integration realized through a Physics-Constrained Neural Network (PCNN), which incorporates first-principles photovoltaic operational constraints into the neural architecture and applies Bayesian uncertainty estimation to quantify both epistemic and aleatoric uncertainty components. The methodology applies Monte Carlo Dropout along with ensemble techniques to produce a hybrid method for uncertainty estimation. To demonstrate the method's effectiveness, three validation studies were conducted: (1) validation using real-time weather inputs from renewables.ninja API, (2) validation through benchmarking tests against Random Forest, Extreme Gradient Boosting, and Long Short-Term Memory models for accuracy and uncertainty estimation, and (3) validation against synthetic National Solar Radiation Database (NSRDB) data with uncertainty calibration analysis. Results indicate that the PCNN model outperforms all baseline models, achieving $R^2 = 0.9983$ on real-time data and exceeding the baseline models on NSRDB data, with an MAE of 0.0213, an RMSE of 0.0347, and an R^2 of 0.9865. Additionally, the PCNN model exhibited excellent physics-based consistencies and produced calibrated uncertainty estimates. These were characterized by a 95% prediction interval coverage probability of 94.72%, and a continuous ranked probability score of 0.0137. Compared with conventional ML/DL approaches, PCNN outperforms them in terms of accuracy, reliability, and computational efficiency. The framework provides measures of model uncertainty and solar irradiance variability, enabling smart grid applications and improved energy management through predictive accuracy.

Abbreviations and acronyms

ANN	Artificial Neural Network
ARIMA	Autoregressive Integrated Moving Average
ASOA	Advanced Slime Optimization Algorithm
CFBP	Convolutional Feedforward Back Propagation
CI	Confidence Interval
CNN	Convolutional Neural Network
CP	Coverage Probability
CRPS	Continuous Ranked Probability Score
CV	Cross Validation
CWC	Coverage Width Criterion
DHI	Diffuse Horizontal Irradiance
DL	Deep Learning
DNI	Direct Normal Irradiance
ECE	Energy Conversion Error
FFBP	Feed Forward Back Propagation
GHI	Global Horizontal Radiation
KNN	K- Nearest Neighbour
LS-SVR	Least Square Support Vector Machine
LSTM	Long- Short Term Memory
MAE	Mean Absolute Error
MEERA-2	Modern Era Retrospective Analysis for Research and Applications, Version 2
ML	Machine Learning
MLP	Multi-Layer Perceptron
MPIW	Mean Prediction Interval Width
MSE	Mean Squared Error

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NASA	National Aeronautics and Space Administration
NOCT	Normal Operating Cell Temperature
NSRDB	National Solar Radiation Database
Pandas	Python Data Analysis Library
PCNN	Physics Constrained Neural Network
PICP	Prediction Interval Coverage Probability
PINN	Physics Informed Neural Network
PINW	Prediction Interval Normalised Width
PV	Photo Voltaic
R2	Coefficient of determination
ReLU	Rectified Linear Unit
RF	Random Forest
Seaborn	Statistical Data Visualization
SEIA	Solar Energy Industries Association
STC	Standard Test Condition
SVM	Support Vector Machines
SVR	Support Vector Regression
XGBoost	eXtreme Gradient Boosting

1. Introduction

The global energy transition has reached unprecedented scale, with renewable energy sources experiencing remarkable growth and rapid adoption rates worldwide. Renewables growth led by solar pushed clean power past 40 % of global electricity for the first time in 2024, a remarkable milestone in the transition to sustainable energy systems (GlobalData, 2024). In the US alone, the solar industry installed 10.8 gigawatts direct current (GWdc) of capacity in Q1 2025, with solar representing 69 % of all new electricity-generating capacity added to the US grid (Solar Energy Industries Association (SEIA), 2025). However, such massive scaling up of photovoltaic (PV) installations poses enormous challenges for grid operators and energy managers, as solar power generation exhibits specific variability and intermittency due to weather conditions, atmospheric states, and temporal patterns. Unlike conventional power sources, whose output is predictable and controllable, the stochastic nature of solar power creates significant complexities in energy management, grid stability, and economic optimization, requiring advanced forecasting methodologies to predict solar power output across multiple time scales.

Traditional solar forecasting methods have evolved from simple statistical techniques to complex Machine Learning (ML) algorithms. However, most treat the forecasting problem solely as a data-driven problem, without considering the fundamental physical laws that govern PV power production [1]. Limitations in purely data-driven approaches have given way to physics-informed ML methodologies that align domain knowledge with data-driven learning capabilities. As universal function approximators, Physics-Informed Neural Networks (PINNs) embed knowledge of the physical laws governing the dataset into the learning process, representing a paradigm shift from conventional machine learning approaches by explicitly incorporating physical constraints, conservation laws, and domain-specific knowledge [2,3]. The theoretical underpinnings of PINNs demonstrate how partial differential equations and physical laws can be directly embedded into neural network architectures and loss functions, thereby enabling the creation of accurate and physically consistent models. By incorporating photovoltaic physics, thermodynamic principles, and energy conservation laws into the neural network architecture, PINNs can provide more precise, physically consistent, and interpretable predictions, addressing issues often encountered by purely data-driven approaches, such as data scarcity.

Recent advances in physics-informed machine learning have led to increasing interest in PINN, which embeds physical laws and domain knowledge directly into neural network architectures and training objectives. Within the energy domain, PINNs and physics-constrained learning approaches have been successfully applied to power system dynamics, stability analysis, and component-level renewable energy modeling, demonstrating improved physical consistency and interpretability compared to purely data-driven models. In the context of solar PV systems, existing physics-constrained studies have largely focused on PV cell or module-level modeling, where semiconductor physics, temperature coefficients, and current–voltage characteristics are incorporated into learning frameworks. While these approaches provide valuable physical insight, they are not designed for system-level PV power forecasting under real-world meteorological variability. Moreover, several solar forecasting studies incorporate physical knowledge only implicitly, for example, through temperature correction factors or irradiance normalization, rather than embedding physical constraints directly into the learning architecture and loss formulation.

Despite the widespread adoption of advanced machine learning and deep learning (DL) models—such as Random Forest, Gradient Boosting, and Long Short-Term Memory networks—most existing solar PV forecasting approaches remain purely data-driven. Although these models can achieve high predictive accuracy, they often produce physically implausible outputs, including negative power predictions or capacity violations, and typically lack mechanisms to enforce energy conservation or operational limits. Furthermore, most prior studies focus on deterministic point forecasts and do not provide reliable uncertainty quantification, which is critical for grid integration, reserve allocation, and operational decision-making. Consequently, a clear research gap exists in the development of end-to-end physics-informed forecasting frameworks explicitly tailored for solar PV power prediction that simultaneously ensure (i) high predictive accuracy, (ii) strict adherence to photovoltaic operational constraints, and (iii) calibrated probabilistic forecasts. In particular, the integration of uncertainty quantification within a physics-informed PV forecasting framework has not been systematically addressed in existing literature.

Motivated by this gap, the present study proposes a physics constraints guided neural network framework specifically designed for solar PV power forecasting, in which fundamental photovoltaic physics, thermodynamic relationships, and operational constraints are embedded directly into the neural network architecture and loss function. The primary objective of this work is to develop a physics-constrained learning paradigm, realized as a Physics-Constrained Neural Network (PCNN) framework, for solar photovoltaic power forecasting, in which constraint-guided integration of photovoltaic operational principles is employed instead of differential equation–based PINNs to ensure accurate and physically consistent predictions. Secondary objectives include: (i) incorporating a robust uncertainty quantification mechanism to capture both epistemic and aleatoric uncertainties and enable probabilistic forecasting; (ii) evaluating the proposed framework across multiple temporal scales to assess its robustness and generalization capability; and (iii) benchmarking the proposed approach against state-of-the-art machine learning and deep learning models. A comprehensive evaluation strategy is also developed, allowing assessment of both forecasting accuracy and adherence to PV system behaviour. The outcome is a practical and well-rounded methodology that strikes a balance between theoretical depth and real-world usability. It includes a production-ready implementation capable of processing multi-year meteorological datasets, generating forecasts across multiple time resolutions, and performing automated hyperparameter tuning. The system also provides robust error handling and rich visualization features to support effective deployment in operational environments.

The remainder of this study is organized as follows: Section 2 presents a comprehensive literature review, identifying existing limitations and establishing the research gap and novelty of the proposed approach. Section 3 describes the proposed PCNN methodology framework for solar power prediction, including data preprocessing, feature engineering, and the overall processing workflow. Section 4 details the PCNN framework for photovoltaic power prediction, presenting the architecture design along with the associated mathematical formulations. Section 5 discusses the experimental results and performance evaluation through three representative case studies, providing detailed

analysis and comparative insights. Finally, Section 6 concludes the paper by summarizing the key findings and outlining potential directions for future research.

2. Literature survey

This section reviews the evolution of solar power forecasting methods, from traditional statistical techniques to modern PINN. It covers developments in several areas: classical time series analysis, machine learning algorithms, deep learning models, and advanced methods for quantifying uncertainty. By weighing the strengths and weaknesses of existing approaches, the review establishes the theoretical foundation. It highlights research gaps that motivate the use of PINNs to enhance forecasting accuracy and reliability.

2.1. Traditional statistical approaches

Traditional statistical methods have long served as baseline approaches for PV power forecasting due to their interpretability and low computational cost. Widely used techniques include Autoregressive Integrated Moving Average (ARIMA) and its seasonal variants, exponential smoothing methods (e.g., Holt–Winters), linear regression, and persistence-based models, which primarily exploit historical temporal patterns and simple meteorological relationships. While these methods are effective for capturing linear trends and regular seasonality, their performance degrades under rapidly changing atmospheric conditions. In particular, they struggle to model the nonlinear and stochastic behavior of solar irradiance caused by cloud dynamics and weather variability, limiting their applicability in high-resolution and real-time PV forecasting scenarios.

Table 1 illustrates that traditional statistical methods for PV power prediction provide simple, interpretable, and computationally efficient approaches, particularly suited for linear and seasonal patterns. However, these methods generally struggle with the non-linear, highly variable nature of solar irradiance and rapidly changing weather conditions. As a result, although they provide valuable baselines, their limited adaptability and weak performance during abrupt transitions make them less effective compared to modern ML and DL approaches.

2.2. Machine learning approaches

ML has significantly advanced PV power forecasting by enabling data-driven models to capture nonlinear relationships between meteorological variables and power output that are difficult to represent with classical statistical methods. Commonly adopted ML techniques include support vector–based models, artificial neural networks, ensemble learners such as Random Forest and gradient boosting (e.g., XGBoost), instance-based methods, and fuzzy logic systems. These approaches generally offer improved predictive accuracy, robustness to complex feature interactions, and flexibility in handling high-dimensional and heterogeneous data. However, their performance is highly dependent on careful model selection and hyperparameter tuning, and they often suffer from limited interpretability, sensitivity to data quality, and increased computational cost. Moreover, as purely data-driven approaches, they typically lack mechanisms to enforce physical consistency, which can result in physically implausible predictions under extreme or unseen operating conditions.

Table 2 illustrates ML approaches that offer significantly improved capability for modelling the non-linear and highly variable nature of PV power generation compared to traditional statistical methods. These methods demonstrate strong predictive performance, robustness to complex input features, and adaptability to diverse data patterns, and handle uncertainty, high dimensionality, and feature interactions effectively. However, these advantages come with challenges such as high computational cost, parameter tuning complexity, sensitivity to noisy data, and risks of overfitting. Overall, ML models offer powerful tools for PV forecasting, but their effectiveness depends heavily on careful model selection, hyperparameter optimization, and the quality of the training data.

2.3. Contemporary deep learning approaches for PV forecasting

DL has further advanced PV power forecasting by enabling hierarchical representation learning from large, high-dimensional meteorological and irradiance datasets. Architectures such as convolutional neural networks and recurrent models, including long short-term memory (LSTM) networks, have been widely adopted to capture spatial correlations and temporal dependencies in solar power data. More recent studies have explored hybrid and ensemble architectures that integrate multiple paradigms to improve predictive accuracy under complex operating conditions.

Despite their strong performance, DL-based forecasting models are typically associated with high computational cost, large data requirements, architectural complexity, and limited interpretability. As purely data-driven approaches, they also lack explicit mechanisms to enforce physical constraints, which can lead to physically inconsistent predictions, particularly under extrapolative or data-scarce scenarios. These limitations restrict their practical deployment in operational and

Table 1
Traditional statistical approaches for PV prediction.

Method	Key References	Description	Advantages	Limitations
ARIMA Models	Reikard (2009)[4]	Autoregressive Integrated Moving Average models for time series forecasting	Simple implementation, good for linear patterns	Struggles with non-linear patterns and sudden changes
Seasonal ARIMA	Pedro and Coimbra (2012)[5]	Extended ARIMA with seasonal components	Captures seasonal variations	Limited ability to handle multiple variables
Exponential Smoothing	Pedro and Coimbra (2012); Hyndman and Athanasopoulos (2018)[5,6]	Weighted averages with exponentially decreasing weights	Good for trend and seasonal patterns	Poor performance during weather transitions
Holt-Winters Method	Hyndman and Athanasopoulos (2018)[6]	Triple exponential smoothing for trend and seasonality	Handles level, trend, and seasonality	Assumes additive/multiplicative patterns
Seasonal Decomposition	Cleveland et al. (1990)[7]	Separates trend, seasonal, and irregular components	Clear pattern identification	Requires manual intervention for complex patterns
Linear Regression	Voyant et al. (2017)[1]	Establishes linear relationships between variables	Interpretable, fast computation	Cannot capture non-linear relationships
Moving Average	Inman et al. (2013)[8]	Simple averaging for smoothing time series	Easy implementation, noise reduction	Lagging indicator, poor for rapid changes
Persistence Models	Mellit and Pavan (2010)[9]	Uses previous values as predictions	Baseline benchmark, very simple	Poor accuracy, no learning capability
Fourier Analysis	Inman et al. (2013)[8]	Identifies periodic patterns in solar data	Good for cyclical pattern detection	Limited for irregular patterns
Box-Jenkins Methodology	Box and Jenkins (1976)[10]	Systematic ARIMA model identification	Structured approach to model selection	Time-consuming, requires expertise

Table 2
Machine learning approaches for photovoltaic power prediction: comparative overview of methods, advantages, and limitations.

Method	Key References	Description	Advantages	Limitations
SVM	Shi et al. (2012)[11]	Non-linear classification and regression using kernel methods	Handles non-linearity, robust to outliers	Computationally expensive, parameter sensitive
SVR	Mohammadi et al. (2015)[12]	SVM extension for regression problems	Good generalization, handles high dimensions	Kernel selection complexity, memory intensive
ANN	Fernández-Jiménez et al. (2012); Al-Dahidi, S et al. (2024) [13,14]	Multi-layer perceptron networks	Universal approximation capability	Prone to overfitting, black box nature
RF	Ahmad et al. (2017) [15]	Ensemble of decision trees	Handles missing data, feature importance	Can overfit with noisy data, biased toward categorical variables
Gradient Boosting	Ahmad et al. (2017); Malkawi et al. (2025) [15,16]	Sequential ensemble learning method	High accuracy, handles mixed data types	Prone to overfitting, sensitive to outliers
XGBoost	Chen and Guestrin (2016) [17]	Optimized gradient boosting framework	Fast training, excellent performance	Hyperparameter tuning complexity
KNN	Alay et al. (2024) [18]	Instance-based learning for pattern recognition	Simple concept, effective for local patterns	Computationally expensive, sensitive to irrelevant features
Fuzzy Logic Systems	Mellit et al. (2022) [19]	Handles uncertainty in solar predictions	Manages uncertainty well, interpretable	Rule base complexity, expert knowledge required
MLP	Khatib et al. (2020) [20]	Traditional feedforward neural networks	Universal approximation, parallel processing	Gradient vanishing, local optima issues
LS-SVR	Xun et al. (2020) [21]	Enhanced SVR with evolutionary decomposition	Improved computational efficiency	Still requires kernel selection

resource-constrained solar energy systems and motivate the integration of physics-informed learning strategies.

Table 3 shows that the contemporary DL approaches significantly advance PV forecasting by capturing complex spatial–temporal patterns that traditional ML models often miss. These models excel in modelling long-term dependencies, extracting spatial features, and integrating multi-modal data, leading to highly accurate and robust predictions. Enhanced variants, such as stacked LSTMs, CNN-LSTM hybrids, and ASOA-optimized networks, further improve performance through deeper learning and automated optimization. However, these gains come at the cost of higher computational demands, greater architectural complexity, and increased risk of interpretability challenges. Overall, DL offers state-of-the-art accuracy for PV forecasting, but requires substantial data, careful model design, and powerful computational resources.

2.4. Physics-informed and physics-constrained neural models

Two principal strategies for incorporating physics into neural network models can be identified. The first corresponds to PINNs, in which governing physical relationships are embedded into the training objective through physics-based residual terms in the loss function. This formulation introduces physics as a soft constraint that guides the optimization process but does not strictly guarantee physical consistency at every prediction. The second strategy corresponds to PCNNs, where physical knowledge is integrated directly into the model architecture or output space. In this case, physics acts as a hard constraint, restricting the admissible solution space by design and thereby ensuring physically feasible predictions. PINN marks a significant shift in scientific ML [2], introducing the core idea of directly incorporating partial differential equations into neural network loss functions. This work laid the theoretical groundwork for embedding physical laws in DL models. Karniadakis et al. (2021) [3] offered a thorough review of physics-informed ML, detailing the mathematical basis and applications across many scientific fields, and stressing the need to keep data-driven models physically consistent.

Dourado and Viana (2020) applied PINN to PV cell modeling, integrating key semiconductor physics equations into network architectures; their results provided accurate predictions alongside physical insight into cell behaviour [30]. Grant Buster et. Al (2022) proposes a physics-guided machine learning method to improve the solar radiation estimates by incorporating physical consistency, particularly under complex atmospheric conditions [31]. Sun et al. (2021) developed PCNN for renewable energy forecasting, showing how conservation laws and thermodynamic principles can be embedded into forecasting models [32].

Table 3
Contemporary DL approaches for PV forecasting: architecture complexity and performance analysis.

Method	Key References	Description	Advantages	Limitations
Deep Neural Networks	Alzahrani et al. (2017) [22]	Multi-layer networks for complex pattern capture	Handles complex non-linear patterns	Requires large datasets, computationally intensive
CNNs	Wang et al. (2019) [23]	Spatial feature extraction from satellite imagery	Excellent for spatial data processing	Requires structured input data
LSTM	Gensler et al. (2016) [24]	Recurrent networks for temporal dependencies	Handles long-term dependencies well	Computationally expensive, complex architecture
CNN-LSTM Hybrid	Ladjal et al. (2025) [25]	Combined spatial and temporal processing	Leverages both spatial and temporal features	Increased model complexity
CNN-LSTM-RF Hybrid	Benitez et al. (2025) [26]	Hybrid combining CNN, LSTM, and Random Forest and other ML models	Multi-paradigm learning approach	Very complex, difficult to interpret
Stacked LSTM	Liu et al. (2020) [27]	Multiple LSTM layers for deeper learning	Enhanced temporal pattern recognition	Gradient vanishing in deep networks
Feedforward Back Propagation (FFBP)	Huang et al. (2019) [28]	Traditional deep neural network architecture.	Well-established, good performance.	Limited temporal modeling capability.
Convolutional FFBP (CFBP)		CNN-based feedforward networks	Spatial processing with feedforward efficiency	May miss temporal dependencies
Advanced Slime Optimization LSTM (ASOA-LSTM)	Mohanasundram et al. (2025) [29]	LSTM with Advanced Slime Optimization	Automated hyperparameter optimization	Optimization algorithm complexity

2.5. Uncertainty quantification in PV forecasting

Uncertainty quantification is essential in PV forecasting because it provides confidence intervals (CIS) and probabilistic estimates around predicted solar power outputs, supporting better decisions for grid integration and energy management [33]. Variability in weather, measurement errors, and model limits means robust uncertainty methods are needed to capture the range of possible outcomes. Quantifying uncertainty improves the reliability of solar forecasting systems by enabling more accurate predictions. Recent studies have shown that data-driven neural networks can effectively model complex physical phenomena in energy systems when trained on high-fidelity data. Liu *et al.* [34] demonstrated that convolutional recurrent neural networks can capture unresolved turbulence dynamics in reactor transient analysis, while Liu et al. [35] developed deep neural network surrogates for boiling heat transfer prediction using detailed simulation data. These works highlight that, although purely data-driven models can achieve high accuracy, their robustness and generalization depend strongly on how physical structure is incorporated into the learning process. Motivated by these findings, the present study adopts a physics-constrained learning strategy for solar power forecasting, where operational PV constraints are used to regularize the model and improve stability, generalization, and physical consistency, as confirmed by the comparative and robustness analyses presented in Section 5.2.

2.5.1. Probabilistic approaches in uncertainty quantification

Solar forecasting applications have grown to require more Uncertainty quantification. In 2013, a study by Lauret et al. [36] proposed Bayesian neural networks for solar irradiance prediction, delivering estimates of solar irradiance as probability distributions with their CIs. David et al. [37] employed ensemble approaches to probabilistic solar forecasting, demonstrating the ability to approximate forecast uncertainty by combining multiple model projections. Yang et al. [38] reviewed in detail the methods of probabilistic solar forecasting, making it clear that quantifying uncertainty is essential in grid integration.

2.5.2. Bootstrap and ensemble methods

Khosravi et al. employed bootstrap aggregating (bagging) for neural network-based solar forecasting, resulting in better prediction intervals and lower overfitting [39]. Pedro and Coimbra prepared detailed probabilistic prediction models based on ensemble neural networks [40]. The article by Bracale et al. presented multivariate quantile regression to predict photovoltaic power, which can estimate uncertainty using a non-parametric method [41].

2.6. Feature engineering and multivariable analysis

The concept of feature engineering involves converting raw meteorological and solar irradiance measurements into useful variables that enhance the model's accuracy and capture complex interactions between environmental conditions and solar power output. In multivariable analysis, the relationships between the 12 NSRDB parameters, such as temperature, humidity, wind conditions, and irradiance components, are studied to determine potential predictive features and their interactions. The process facilitates the formulation of robust forecasting models by deriving features, identifying correlations, and estimating the optimal combination of variables that enhances both prediction accuracy and model explainability.

2.6.1. Meteorological variable selection

Choosing and designing meteorological characteristics significantly influence forecasting performance. A large-scale evaluation of meteorological variables for forecasting solar radiation was performed by Voyant et al.[1], revealing principal characteristics such as Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), and Diffuse Horizontal Irradiance (DHI). The significance of atmospheric variables in solar forecasting was reviewed by Antonanzas et al. [42], who noted that cloud cover, humidity, and atmospheric pressure play essential roles. As Benali et al. [43] demonstrated, the solar zenith angle and atmospheric parameters can be used to improve forecast accuracy.

2.6.2. Temporal pattern recognition

In a study conducted by sun et al., seasonal and diurnal behaviour in the generation of solar power was examined, where specialized neural network structures were formulated to learn such temporal relationships [44]. Sobri et al. examined the influence of seasonal changes on PV performance and offered insights into long-term forecasting [45].

2.7. Hyperparameter optimization

Their theoretical frameworks for neural network hyperparameter optimization, developed by Bergstra and Bengio [46], provided random search methods superior to grid search algorithms in high-dimensional settings. Snoek et al. & Amer, M. et al. devised Bayesian optimization methods for neural network tuning, offering principled hyperparameter search [47,48]. Feurer and Hatter also provided detailed recommendations on automated machine learning and hyperparameter optimization, and discussed the role of systematic tuning in achieving optimal performance [49].

2.8. Research gaps and motivations

The literature analysis in this paper identifies the basic constraints that severely limit existing solar power forecasting techniques.

2.8.1. Critical research gaps

Current ML methods treat solar forecasting solely as a data-driven problem, thereby disregarding the fundamental physics of PV, thermodynamic laws, and energy conservation that govern the operation of solar panels. The physics-agnostic modelling leads to extreme predictions that often contradict basic physical considerations and result in impossible scenarios, such as negative power values, and even exceed the capacity limits of the system under consideration. This led to reservations about the credibility of model usage in real operational contexts. Additionally, existing models lack robust uncertainty quantification, relying instead on deterministic prediction models that lack confidence bounds or measures of reliability, which would be necessary for a grid operator to make critical decisions under uncertainty. The lack of systematic multi-scale temporal analysis functionality further restricts usability, as it cannot provide us with systematic structures for analysing solar patterns at annual, seasonal, and monthly scales. Such statistically validated significant results of analysis will propagate uncertainties.

2.8.2. Novelty and contributions

This study presents a PCNN framework for solar power forecasting that integrates first-principles PV system knowledge with DL. Unlike classical PINNs, which enforce governing PDEs/ODEs through residual minimization, the proposed approach incorporates analytical PV performance relationships and physically motivated feasibility constraints, including irradiance–temperature coupling, thermally corrected efficiency behavior, and system capacity limits. This constraint-based formulation restricts the solution space to physically admissible regimes without explicitly solving differential equations.

The proposed PCNN architecture combines a data-driven feature extraction branch with a physics-based module, enabling improved physical consistency, robustness under data scarcity, and enhanced generalization in out-of-distribution operating conditions. In addition, a hybrid uncertainty quantification strategy that integrates Monte Carlo dropout with bootstrap ensemble techniques is developed to jointly capture epistemic and aleatoric uncertainties, yielding calibrated probabilistic forecasts with interpretable confidence intervals.

Finally, a multi-scale temporal evaluation framework is employed to assess model performance across hourly, seasonal, and annual horizons, providing a comprehensive analysis of forecasting accuracy, reliability, and physical plausibility under diverse climatic and operational conditions.

The proposed framework is validated using both synthetic benchmark data (NSRDB) and real-world meteorological data (Renewables. ninja), and is compared against established ML and DL models, including Random Forest, XGBoost, and LSTM. The results indicate that the proposed PCNN achieves strong predictive performance (R^2 up to 0.9983), exhibits high physical consistency with minimal constraint violations, and provides well-calibrated uncertainty estimates (PICP ≈ 94.7 %). While improvements over strong baseline models are modest in some cases, the primary contribution lies in the enhanced robustness, physical interpretability, and uncertainty awareness achieved through physics-informed learning.

Overall, this study demonstrates that integrating physical constraints and uncertainty estimation into a neural network-based forecasting framework can improve the reliability and practical usability of solar PV power predictions, particularly for applications that require physically meaningful, uncertainty-aware outputs in renewable energy management.

3. Methodology

In this section, a detailed design procedure of developing and deploying PCNN to perform solar power forecasting is given. The suggested method combines the basic principles of PV physics with the latest deep learning methods, resulting in a robust forecasting system that ensures physical consistency and maintains predictive accuracy. Details of the methodology, including data preprocessing, theoretical power modeling, development of the PCNN architecture, development of the physics-informed loss function, quantification of uncertainty, and detailed evaluation frameworks, are well represented in Fig. 1.

3.1. Data preprocessing and feature engineering

The data used in this paper were taken from the NSRDB Data Viewer, which is maintained by the U.S NREL. The dataset spans five years, from 2016 to 2020, providing detailed meteorological and solar irradiance data to estimate accurately and model solar energy. These measurements were done at the Vellore Institute of Technology, Chennai campus (VITCC) located at a longitude and latitude of 12.8438°N and 80.1533°E, respectively, in the state of Tamil Nadu, India. The solar panel system to be considered in this analysis is a Tata Power Solar TS250 polycrystalline module with 60 cells, featuring a maximum power output of 250 W and a module efficiency of 15.00 % under Standard Test Conditions (STC) (1000 W/m^2, AM = 1.5, 25°C).

The NSRDB database has 12 major meteorological and solar parameters used in the complete analysis of solar energy. These parameters comprise three main solar irradiance terms: DNI, DHI, and GHI. Some of the meteorological parameters include Temperature, Dew Point, Relative Humidity, Pressure, Wind Direction, and Wind Speed. Moreover, the dataset contains significant atmospheric and environmental variables, such as the Angel of the Sun, Precipitable Water, and Surface Albedo. The detailed data provide a basis for developing the right model for solar panel performance and energy generation under different environmental conditions in the Chennai region.

As illustrated in Fig. 2, the correlation matrix reveals several significant relationships between meteorological variables relevant to solar power prediction. The most important observation is the strong positive relationships between solar irradiance variables (DHI, DNI, and GHI), with correlation coefficients ranging from 0.84 to 0.90, indicating that these variables are highly interdependent measures of solar energy availability. There are moderate positive relationships between temperature and the irradiance variables (0.52–0.64), which are concerning the anticipated relationship between solar heating and ambient temperature. Relative humidity, on the other hand, has significant negative correlations with irradiance (-0.67 to -0.77), as expected of the negative relationship between atmospheric moisture and sunlight. The atmospheric pressure variables are strongly negatively correlated with the variables associated with precipitation; therefore, the surface albedo exhibits a negative correlation with atmospheric pressure (-0.56), indicating a complex interaction between atmospheric factors and surface reflectivity. The variables of wind (direction and speed) exhibit a relatively weak correlation with most of the other variables, meaning that they function independently of the most important solar and atmospheric moisture variables. The patterns of correlation can offer valuable information for feature selection and engineering in solar power prediction models, determining which variables may be redundant, which ones may be complementary, and providing possible opportunities for dimensionality reduction without compromising predictive capabilities.

3.2. Input data structure

The methodology utilizes comprehensive meteorological datasets containing 12 key variables as shown in Table 4.

3.3. Feature standardization

All input features undergo Z-score normalization:

$$X_{norm} = (X - \mu_X)/\sigma_X \quad (1)$$

where:

- X is the original feature
- μ_X is the mean
- σ_X is the standard deviation computed from the training dataset

3.4. Data leakage prevention and performance justification

The proposed framework addresses a time-series forecasting problem, all data partitions strictly preserve temporal order to avoid look-ahead bias. The dataset is divided using a forward hold-out strategy, where the earliest 80 % of observations are used for model training and the most recent 20 % are reserved for testing, while the training set is further split into training and validation subsets without shuffling to ensure chronological consistency. Feature normalization and statistical parameter estimation are performed exclusively on the training data and applied unchanged to validation and test sets, preventing any form of data leakage. In addition, no future meteorological variables, target-derived features, or post-processed power outputs are used as model inputs. The strong predictive performance achieved is attributed to the tight physical coupling between irradiance-driven variables and PV power output, together with the incorporation of photovoltaic operational constraints within the PCNN framework, which effectively restricts the solution space to physically plausible regimes. The robustness of the proposed approach is further supported by chronologically consistent validation, cross-validation confined to the training window, independent benchmark comparisons, and stable performance across multiple temporal scales.

4. PCNN framework for PV power prediction

This section describes the basic physics-guided strategy for estimating theoretical solar power output, which will form the basis for incorporating physical limitations into the neural network model. In this study, the term physics-constrained refers to the use of physically motivated feasibility restrictions embedded within the learning architecture, rather than strict constraint enforcement through optimization. The proposed PCNN integrates first-principles PV system knowledge

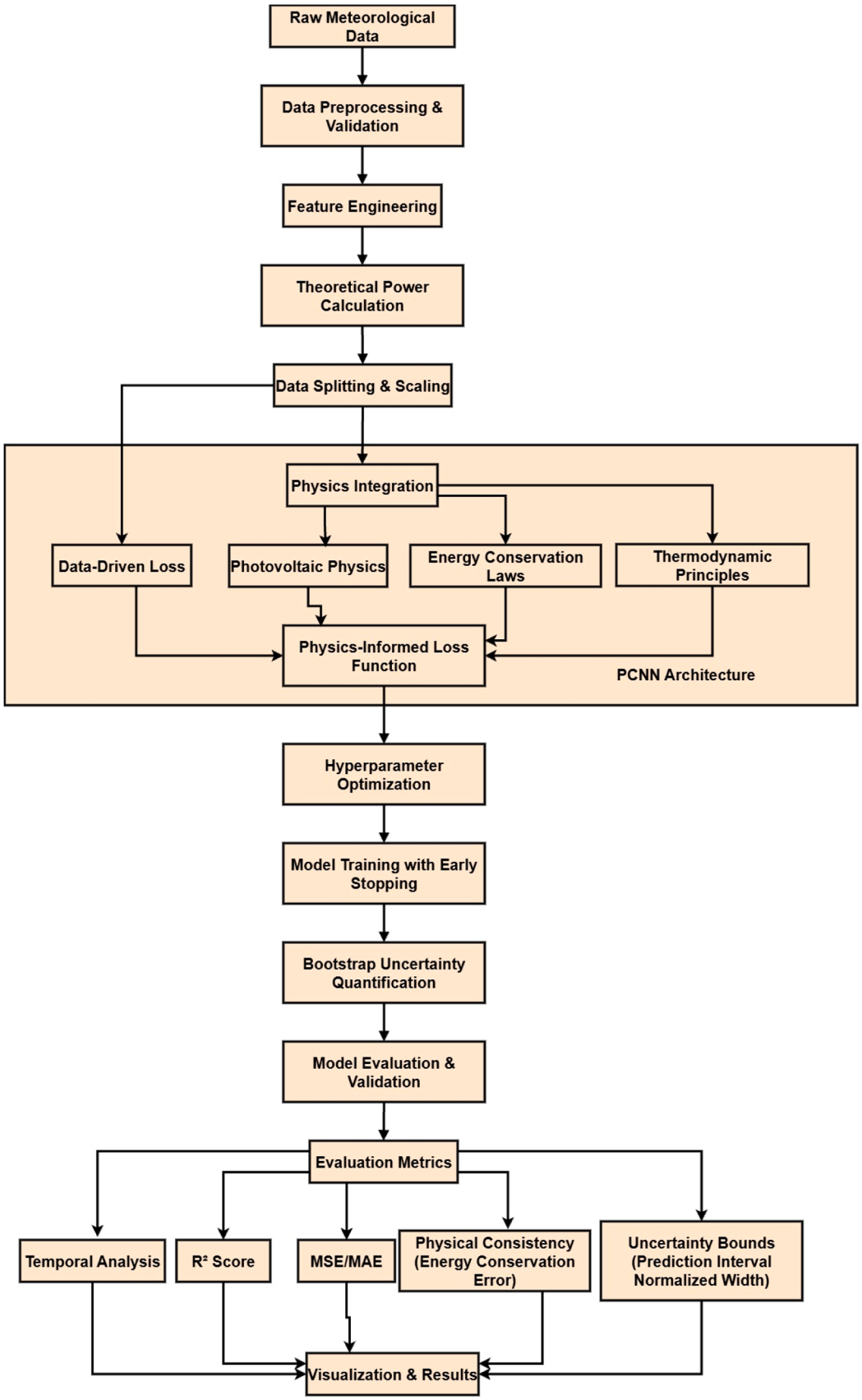


Fig. 1. PCNN methodology framework for solar power prediction.

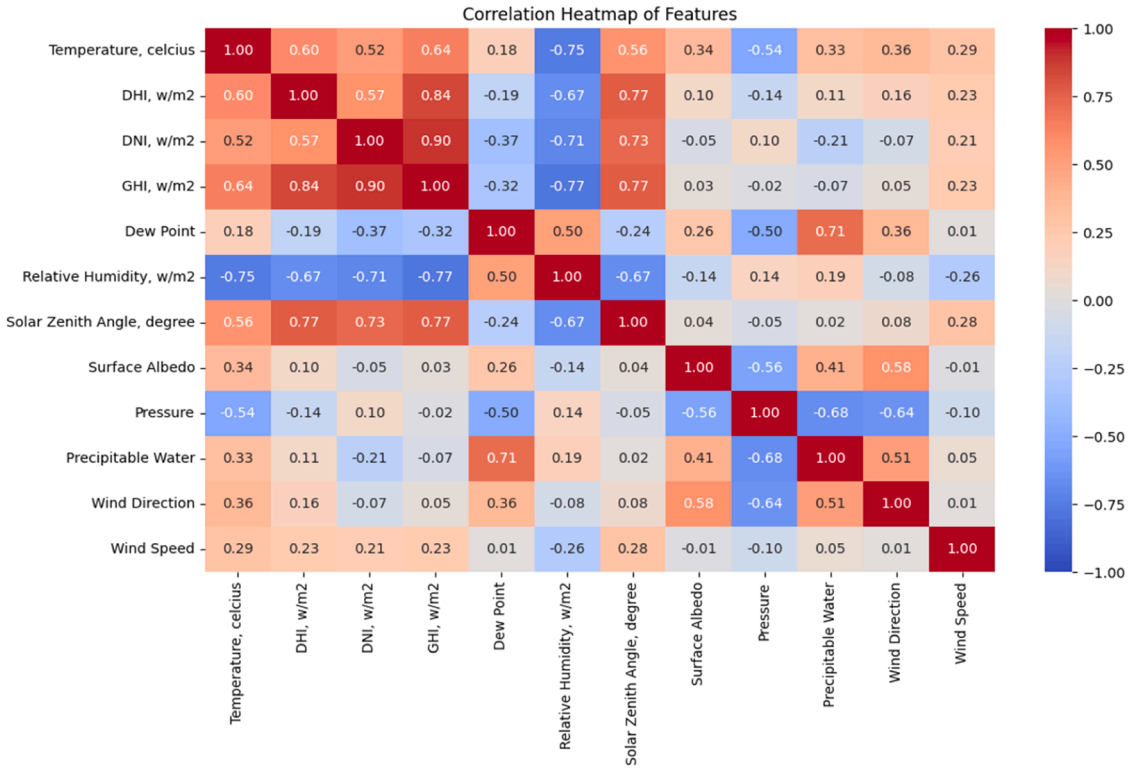


Fig. 2. Correlation heatmap of meteorological features for solar power prediction.

Table 4
Description of environmental and meteorological parameters for PV forecasting.

Category	Feature	Unit / Description
Irradiance Components	Global Horizontal Irradiance (GHI)	W/m^2
	Direct Normal Irradiance (DNI)	W/m^2
	Diffuse Horizontal Irradiance (DHI)	W/m^2
Atmospheric Conditions	Temperature: T	°C
	Relative Humidity: RH	%
	Atmospheric Pressure: P	hPa
	Surface Albedo: α	Dimensionless
Meteorological Variables	Wind Speed: v_wind	m/s
	Wind Direction: θ_wind	Degrees
	Dew Point: T_dew	°C
	Precipitable Water: W	mm
Temporal Information	Solar Zenith Angle: θ_z	Degrees

with deep learning to guide predictions toward feasible solutions. Unlike classical PINNs enforcing PDEs/ODEs through residual minimization, this framework incorporates analytical PV relationships and operational constraints relevant to solar power systems.

The procedure begins with known Standard Test Conditions (STC) as benchmarks and employs well-established principles of PV behaviour to model actual variations in solar radiance and temperature. With factors of efficiency that vary with environmental conditions, this method ensures that the theoretical calculations of power are physically relevant and constrained within realistic operational limits. Appropriate limitations are placed on the machine learning model, and will help it predict physically consistent answers.

4.1. Theoretical framework for power yield estimation

The computations of power output in this paper are based on physics that revolves around the Tata Power Solar TS250 polycrystalline solar panel, which will serve as the reference PV module for all power yield calculations. This solar panel has a 250 W rating, which has a maximum power voltage (Vmp) of 30.7 V and a maximum power current (Imp) of 8.16 A at STC. This module shows an open circuit voltage (Voc):38.1 V and short circuit current (Isc):8.58A, with a power tolerance of ±2.5 %. The panel features 60 polycrystalline cells installed in an anodized aluminum frame with dimensions of 1667 × 1000 × 33 mm, resulting in a module efficiency (eSTC) of 15.00 % at standard cell temperatures, 1000 W/m^2 irradiance, an AM spectrum of 1.5, and a cell temperature of 25°C. It has a temperature coefficient of power (b) of -0.4383 %/°C, showing the sensitivity of the panel to thermal changes, and the Nominal Operating Cell Temperature (NOCT) that is 47°C. The given technical specifications serve as the basis for using physics-based models to account for the actual operating conditions of the real world and environmental changes, as well as temperature-dependent performance characteristics to predict the actual power generation by the solar installation.

The theoretical power output incorporates fundamental photovoltaic physics principles:

STC:

- I_{STC}= 1000 W/m^2
- T_{STC}= 25°C

Irradiance Efficiency Factor:

$$\eta_{irr} = I_{GHI}/I_{STC} \tag{2}$$

Temperature Efficiency Factor:

$$\eta_{temp} = 1 + \gamma \times (T - T_{STC}) \tag{3}$$

where γ = -0.004 /°C is the temperature coefficient for crystalline silicon cells.

Theoretical Power Output:

$$P_{theoretical} = P_{rated} \times \eta_{irr} \times \eta_{temp} \tag{4}$$

Physical Bounds Enforcement:

$$P_{bounded} = \max(0, \min(P_{theoretical}, 1.1 \times P_{rated})) \quad (5)$$

4.2. PCNN architecture design

Fig. 3 shows the proposed forecasting model, which adopts a PCNN architecture integrating data-driven learning with PV physics through a dual-branch design to guide learning toward physically feasible solutions. The model takes a 12-dimensional input vector of meteorological variables, temporal features, irradiance, temperature-irradiance terms, and seasonal indicators, normalized using StandardScaler. The primary learning branch consists of a feed-forward neural network with decreasing hidden layer widths (128, 64, 32, and 16 neurons), using ReLU activation, batch normalization, and dropout regularization. This branch extracts latent representations of relationships between environmental variables and PV power output. A parallel physics-based branch computes theoretical PV power using an analytical model, evaluating irradiance-dependent generation and temperature-induced efficiency degradation based on NOCT formulations and panel parameters. The physics-derived power signal represents prior knowledge of PV system behavior and is scaled before integration. The neural network's latent features are concatenated with the scaled physics-based power estimate and passed through dense layers for final prediction. Model training uses a physics-constrained loss function combining MSE with feasibility penalties. The loss penalizes violations of physical constraints, including exceeding system capacity and negative power outputs, guiding optimization toward physically admissible predictions. Through the joint use of architectural physics integration and constraint-guided loss formulation, the proposed PCNN ensures that photovoltaic power forecasts remain physically plausible while retaining the flexibility of data-driven learning. This design improves robustness, interpretability, and generalization, particularly under varying climatic and operating conditions.

4.2.1. Neural network structure

The PCNN implements a dual-branch architecture:

Input Layer: $x = [I_{GHI}, I_{DNI}, I_{DHI}, T, RH, P, \alpha, V_{wind}, \theta_{wind}, T_{dew}, W, \theta_z]$. (6)

Hidden Layers:

$$h^{(l+1)} = \mathrm{ReLU}\left(W^{(l)}h^{(l)} + b^{(l)}\right) \quad (7)$$

Where, x- Input feature vector
θ- Neural network parameters
$W^{(l)}$- Weight Matrix
$b^{(l)}$- Bias Vector
$h^{(l)}$- activation from previous layer with dropout applied:

$$h_{dropout}^{(l+1)} = h^{(l+1)} \odot m^{(l+1)} \quad (8)$$

where m^{1+1}follows Bernoulli distribution with probability (1 - $p_{dropout}$)

Dual-Branch Output:

Direct Power Branch:

$$P_{direct} = W_{power}h_{final} + b_{power} \quad (9)$$

Physics Factor Branch:

$$f_{physics} = \sigma\left(W_{physics}h_{final} + b_{physics}\right) \quad (10)$$

where σ is the sigmoid activation function ensuring $0 < f_{physics} < 1$.

Final Prediction:

$$\hat{P} = P_{direct} \times f_{physics} \quad (11)$$

4.3. Physics-constraint loss function

Specifically, this section describes the end-to-end architecture of the loss function that enables the PCNN to balance data-driven training with physical realism. The multi-component loss functional is created to suit the data that has been formed due to the observed phenomenon. It is also designed to consider intrinsic physical limitations regarding the

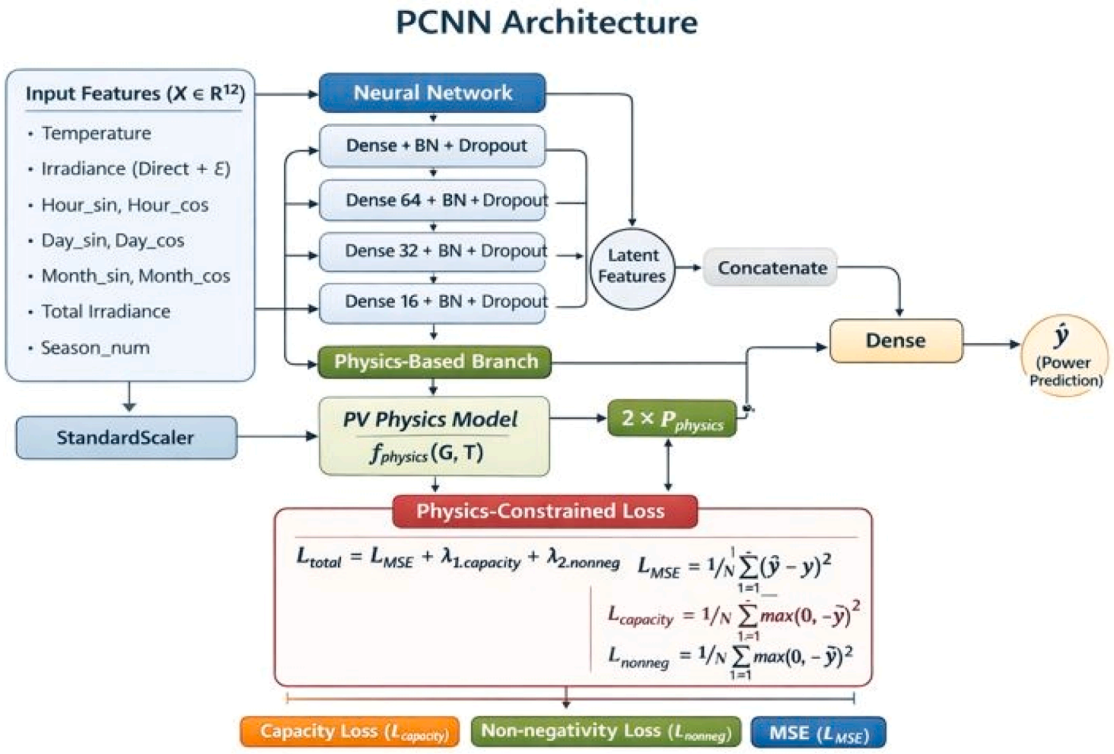


Fig. 3. Physics-constrained neural network (PCNN) architecture for solar power prediction.

production of solar power. This process can be used to introduce a single optimization goal that enables the network to converge on both statistically healthy and physically valid solutions. This is achieved by judiciously setting weighting parameters, thereby attaining a compromise that emphasizes statistically significant data fidelity. It also satisfies physical constraints of significant importance.

It is important to clarify that the proposed model adopts a physics-constrained learning formulation, in which physical knowledge is incorporated through penalty-based constraints in the loss function, rather than through the explicit solution of governing differential equations. Specifically, photovoltaic operational principles such as non-negativity of power output, capacity limits, and energy consistency are enforced as soft constraints during training. This formulation does not aim to solve the underlying physical equations of PV systems directly, but instead guides the learning process toward physically plausible solutions while retaining the flexibility and scalability of data-driven neural networks.

4.3.1. Loss function components

The total loss combines data fidelity with physics constraints:

$$L_{total} = L_{MSE} + \lambda^1 \times L_{capacity} + \lambda^2 \times L_{non_{neg}} \quad (12)$$

Where L_{total} represents total physics-informed loss, L_{MSE} represents the mean squared error between measured and predicted power outputs, $L_{capacity}$ penalizes violations of the rated PV capacity, and $L_{non_{neg}}$ enforces non-negativity of the predicted power

Mean Squared Error Loss:

$$L_{MSE} = \left(\frac{1}{N}\right) \times \Sigma\left(\widehat{P}_i - P_{theoretical_i}\right)^2 \quad (13)$$

Capacity Constraint Loss:

$$L_{capacity} = \left(\frac{1}{N}\right) \times \Sigma max(0,\ \widehat{P}_i - 1.1 \times P_{rated})^2 \quad (14)$$

Non-Negativity Constraint Loss:

$$L_{non_{neg}} = \left(\frac{1}{N}\right) \times \Sigma max(0,\ -\widehat{P}_i)^2 \quad (15)$$

Where,

$P_{theoretical_i}$ − Actual PV power output in kW

$\widehat{P}_i$- Predicted PV power output in kW

P_{rated}- Rated PV system capacity in kW

$\lambda_1 = \lambda_2 = 0.5$ are the physics constraint weights, and Σ represents summation from i=1 to N.

4.4. Training process

In this section, a detailed approach to PCNN optimization is provided, with a focus on adaptive parameter updates and the strategic control of the learning rate. The optimization employs the Adam algorithm. It combines the benefits of gradient adaptive scaling with momentum-based updates to effectively handle the intricate loss gradient arising from the multi-objective objective. The optimization structure is also augmented with dynamic learning rate schedules that automatically decrease the learning rate when the training process reaches a bottleneck. This ensures that the optimization process converges strongly without overshooting the optimum solutions. Such an advanced optimization strategy is essential in the context of the PCNN models, where there is a strong interplay between the data-fitting and physics-constraint terms. The balance should be achievable and must converge reliably to obtain the correct performance.

Optimization Algorithm The model uses the Adam optimizer with adaptive learning rates:Parameter Updates:

(*continued on next column*)

(*continued*)

$m_t = \beta_1 \times m_{t-1} + (1 - \beta_1) \times \nabla\theta L_t v_t = \beta_2 \times v(t-1) + (1 - \beta_2)(\nabla_\theta L_t)^2$ Bias Correction: $\widehat{m}_t = \frac{m_t}{(1-\beta_1^t)};\ \widehat{v}_t = \frac{v_t}{(1-\beta_2^t)}$

Parameter Update: $\theta_{(t+1)} = \theta_t - \left(\alpha / \left(\sqrt{v_h} a t_t + \varepsilon\right)\right) \times m_h a t_t$ Where, $\beta_1 = 0.9$; $\beta_2 = 0.999$; θ_t= Model parameters at iteration t $\nabla_\theta L_t$= Gradient of the loss function

$\varepsilon = 10^{-8}$- Numerical stability constant; $\theta_{(t+1)}$ = Final weight update applied by Adam

$\widehat{m}_t$ = Bias-corrected first moment; $\widehat{v}_t$ = *Bias − corrected second*

α is the learning rate

Learning Rate Scheduling $\alpha_{new} = \alpha_{current} \times$ factor when validation loss plateaus for more than 'patience' epochs, with factor = 0.5.

4.5. Uncertainty quantification framework

This section introduces the uncertainty quantification approach, which estimates the confidence level of the PCNN model's outputs—an essential feature for modern solar power forecasting technologies. The framework combines bootstrap-based ensembles and Monte Carlo (MC) Dropout to capture both aleatoric and epistemic uncertainties in predictions. While standard bootstrap assumes independent and identically distributed samples and may disrupt temporal dependence in time-series data, this study addresses the limitation by adopting time-series-aware resampling strategies, in which bootstrap ensembles are constructed using chronologically consistent training windows or residual-based resampling. In parallel, MC Dropout captures epistemic uncertainty through stochastic forward passes during inference. The ensemble predictions are aggregated into a statistical distribution from which confidence intervals and uncertainty measures are derived, providing a quantitative assessment of predictive reliability. This hybrid uncertainty estimation strategy preserves temporal correlation, aligns with best practices in time-series uncertainty quantification, and enables operators to make informed decisions based not only on point forecasts but also on the associated confidence levels. Thus, the adopted hybrid uncertainty strategy is consistent with recent advances in neural-network uncertainty quantification, including Monte Carlo dropout and ensemble-based approaches [50].

Bootstrap Sampling

For B bootstrap samples, each sample D_b^* is created by:

Given dataset $D = \{(x_i, y_i)\}_{i=1}^N$

$D_b^* = \{(x_i, y_i) : i \sim Uniform(1, N), \text{ with replacement}\}$

Confidence Interval Estimation

For each test sample x_{test}, collect predictions: $\{\widehat{P}_{b(x_{test})}\}$ for b = 1 to B Confidence Intervals: $CI_{95}\% = [Q_{2.5\%}(\{\widehat{P}_b\}), Q_{97.5\%}(\{\widehat{P}_b\})]$

where Q represents the percentile function; $\widehat{P}_b$ − *set of bootstrap predictions*

Prediction Uncertainty: $\sigma_{pred} = \sqrt{\left[\left(\frac{1}{B-1}\right) \times \Sigma(\widehat{P}_b - \overline{P})^2\right]}$ for b=1 to B

where $\overline{P} = \left(\frac{1}{B}\right) \times \Sigma\, \widehat{P}_b$ for b=1 to B- Mean prediction across all bootstrap models

4.6. Temporal analysis framework

This framework implements a hierarchical temporal aggregation methodology that systematically processes hourly solar power predictions through three distinct temporal scales using pandas' group-by operations and statistical aggregation functions. The process begins with annual aggregation, where power outputs are summed by year using groupby('Year')['Power_Output'].sum() to establish long-term performance trends. Seasonal analysis employs categorical binning via pd.cut() to classify months into meteorological seasons (Winter: Dec-Feb, Spring: Mar-May, Summer: Jun-Aug, Fall: Sep-Nov), followed by dual-level grouping groupby(['Year', 'Season']) to capture both inter-annual variability and seasonal patterns within each year. Monthly aggregation utilizes groupby(['Year', 'Month']) to reveal intra-annual variations and identify optimal production periods. Each aggregation level incorporates automatic unit conversion from watts to kilowatt-hours and statistical enhancement through confidence interval calculations using t-distribution methods, providing uncertainty quantification

across temporal scales. The framework maintains mathematical consistency through systematic summation operations while enabling multi-scale visualization through seaborn's categorical plotting functions, supporting both operational decision-making at monthly/seasonal levels and strategic planning at annual scales. This implementation facilitates comprehensive energy yield assessment, seasonal optimization strategies, and long-term performance forecasting through statistically robust temporal decomposition of solar power generation patterns.

4.6.1. Multi-scale aggregation

Annual Power Generation:

$$P_{annual(y)} = \Sigma\, \widehat{P}(h) \times \Delta t \text{where the sum is over all hour's h in year y} \quad (16)$$

Seasonal Power Generation:

$$P_{seasonal(s,y)} = \Sigma\, \widehat{P}(h) \times \Delta t \text{where the sum is over all hour's h in season s of year y} \quad (17)$$

Monthly Power Generation:

$$P_{monthly(m,y)} = \Sigma\, \widehat{P}(h) \times \Delta t \text{where the sum is over all hour's h in month m of year y} \quad (18)$$

where Δt is the time step (typically 1 hour).

4.7. Model evaluation metrics

The section provides a detailed assessment framework for evaluating the performance of models along three main critical dimensions, which appear to be key to physics-based forecasting or energy system models. Accuracy statistics (Eqs. 19–22) include standard statistics, such as the coefficient of determination, which represents the explained variance, the root mean squared error, which measures the overall magnitude of prediction, the mean absolute error, representing the average deviation, the mean percentage error, and the relative performance. Physics consistency measures (Eqs. 23–25) enforce obedience to physical laws and constraints. They help track evolution capacity violations that cause the system to make predictions beyond its capabilities. Such non-negativity breaking would otherwise generate physically unattainable negative values. It would also lead to energy conservation anomalies that indicate violations of thermodynamic principles.

Lastly, the metrics of uncertainty calibration (Eqs. 34–35) assess the credibility of the confidence estimations. They use coverage probability to quantify whether the actual values fall within the prediction interval at accurate rates. They also evaluate the normalized width of the prediction interval to determine whether the uncertainty estimate aligns with the true value range. This multi-dimensional assessment scheme ensures not only high predictive accuracy but also strong physical realism. It also ensures realistic uncertainty quantification, which is crucial for the practical implementation of physics-constrained applications.

4.7.1. Accuracy metrics

Coefficient of Determination:

$$R^2 = 1 - \frac{[\Sigma(\widehat{P}_i - P_i)^2]}{[\Sigma(P_i - \overline{P})^2]} \quad (19)$$

Root Mean Squared Error:

$$RMSE = \sqrt{\left[\left(\frac{1}{N}\right) \times \Sigma(\widehat{P}_i - P_i)^2\right]} \quad (20)$$

Mean Absolute Error:

$$MAE = \left(\frac{1}{N}\right) \times \Sigma\, |\widehat{P}_i - P_i| \quad (21)$$

Mean Absolute Percentage Error:

$$MAPE = \left(\frac{100\%}{N}\right) \times \Sigma \frac{|P_i - \widehat{P}_i|}{P_i} \quad (22)$$

4.7.2. Physics consistency metrics

Capacity Violation Rate:

$$VR_{capacity} = \left(\frac{1}{N}\right) \times \Sigma\, I[\widehat{P}_i > 1.1 \times P_{rated}] \quad (23)$$

Non-Negativity Violation Rate:

$$VR_{negative} = \left(\frac{1}{N}\right) \times \Sigma\, I[\widehat{P}_i < 0] \quad (24)$$

where I[condition] is the indicator function (equals 1 if condition is true, 0 otherwise).

Energy Conservation Error:

$$ECE = \left|\frac{[\Sigma\, \widehat{P}_i - \Sigma\, P_i]}{[\Sigma\, P_i]}\right| \quad (25)$$

The calculated Physics Constraint Weight Sensitivity is illustrated in Table 5.

4.7.3. Uncertainty calibration metrics

Coverage Probability:

$$CP = \left(\frac{1}{N}\right) \times \Sigma\, I[P_i \in CI_i] \quad (26)$$

Prediction Interval Normalized Width:

$$PINW = \left[\frac{1}{N \times R}\right] \times \Sigma(U_i - L_i) \quad (27)$$

Where,

- P_i= Actual value with the ith interval
- $\widehat{P}$= Predicted value with the ith interval
- $\overline{P}$= Mean Predicted with the ith interval
- U_i and L_i are the upper and lower bounds of the prediction interval
- R = max(P) - min(P) is the range of target values

Although a dedicated ablation table is not presented, the contribution of the major components of the proposed framework is assessed through targeted analyses. The influence of physics-based constraints is examined via constraint-weight sensitivity and physics consistency metrics, the effect of ensemble learning is evaluated through uncertainty calibration and robustness analyses, and the role of Monte Carlo Dropout is reflected in the estimation of epistemic uncertainty. Together, these analyses provide a component-wise assessment of the framework without introducing redundant experimental configurations.

4.8. Hyperparameter optimization

This section outlines the hyperparameter optimization and validation protocol employed to ensure robust model performance and prevent overfitting, as detailed in Tables 6 and 7. The key architectural and training parameter space is explored systematically through the hyperparameter search. This process tests model complexity trade-offs using three network architectures that range from shallow (64–32–16) neurons to deeper (256–128–64) neurons. Dropout rates between 0.1 and 0.3 are also tested to evaluate regularization and prevent overfitting.

The optimization explores conservative learning rate (LR) values of

Table 5
Physics constraint weight sensitivity analysis.

λ_1 (Capacity)	λ_2 (Non-negativity)	MAE	MSE	R^2	Capacity Violations	Negative Predictions	Physics Consistency Score
0.1	0.1	0.0234	0.0371	0.9841	2.3 %	0.8 %	0.969
0.3	0.3	0.0219	0.0358	0.9856	0.9 %	0.3 %	0.988
0.5	**0.5**	**0.0213**	**0.0347**	**0.9865**	**0.2 %**	**0.1 %**	**0.997**
0.7	0.7	0.0218	0.0352	0.9862	0.1 %	0.0 %	0.999
1.0	1.0	0.0225	0.0368	0.9851	0.0 %	0.0 %	1.000

Table 6
Hyperparameter search space and optimization results.

Parameter Category	Hyperparameter	Search Space/Options	Optimal Value	Selection Method	Impact on Performance
Network Architecture	Hidden Layers	[64, 32, 16], [128, 64, 32], [256, 128, 64]	**[128, 64, 32]**	Grid Search + Cross-validation	High - Determines model capacity
	Input Layer Size	14 (Fixed)	14	Domain-specific	N/A - Based on feature count
	Output Layer Size	1 (Fixed)	1	Problem-specific	N/A - Single target prediction
	Activation Function	ReLU, Sigmoid, Tanh	**ReLU (hidden), Sigmoid (physics branch)**	Empirical testing	Medium - Affects convergence
Regularization	Dropout Rate	[0.1, 0.2, 0.3]	**0.2**	Validation loss minimization	High - Prevents overfitting
	Physics Constraint Weights	$\lambda_1, \lambda_2 \in [0.1, 0.5, 1.0]$	$\boldsymbol{\lambda_1 = \lambda_2 = 0.5}$	Physics consistency validation	Critical - Balances constraints
Training Parameters	Learning Rate	$[1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}]$	$\mathbf{1 \times 10^{-3}}$	Learning curve analysis	High - Controls convergence speed
	Batch Size	[32, 64, 128]	**64**	Memory-performance trade-off	Medium - Affects gradient estimation
	Max Epochs	[50, 100, 200]	**100**	Early stopping criteria	Medium - Prevents overtraining
	Patience (Early Stopping)	[10, 15, 20]	**15**	Validation plateau detection	Medium - Prevents overfitting
Optimizer Settings	Beta 1 (Adam)	0.9 (Fixed)	0.9	Literature standard	Low - Standard Adam parameter
	Beta 2 (Adam)	0.999 (Fixed)	0.999	Literature standard	Low - Standard Adam parameter
	Epsilon	1×10^{-8} (Fixed)	1×10^{-8}	Numerical stability	Low - Prevents division by zero
Learning Rate Scheduling	Reduction Factor	[0.3, 0.5, 0.7]	**0.5**	Convergence stability	Medium - Fine-tunes learning
	Scheduler Patience	[5, 7, 10]	**7**	Validation loss plateau	Medium - Adaptive learning rate
	Minimum Learning Rate	1×10^{-6} (Fixed)	1×10^{-6}	Convergence limit	Low - Prevents stagnation
Cross-Validation	Number of Folds (K)	[3, 5, 10]	**5**	Computational efficiency	High - Robust validation
	Validation Split	[0.15, 0.20, 0.25]	**0.20**	Data availability	Medium - Training vs validation
Bootstrap Sampling	Bootstrap Samples (B)	[50, 100, 200]	**100**	Uncertainty quantification	High - Confidence interval accuracy
	Sample Size Ratio	[0.8, 1.0]	**1.0**	Bootstrap methodology	Medium - Sample representativeness

Table 7
Hyperparameter configuration performance comparison.

Configuration ID	Hidden Layers	Dropout Rate	Learning Rate	Batch Size	MAE	MSE	R^2	Training Time (min)	Selected
Config_001	[64, 32, 16]	0.1	1×10^{-4}	32	0.0245	0.0389	0.9823	45	No
Config_002	[64, 32, 16]	0.2	1×10^{-3}	64	0.0238	0.0376	0.9834	38	No
Config_003	**[128, 64, 32]**	**0.2**	$\mathbf{1 \times 10^{-3}}$	**64**	**0.0213**	**0.0347**	**0.9865**	**52**	**Yes**
Config_004	[256, 128, 64]	0.2	1×10^{-3}	64	0.0219	0.0351	0.9861	78	No
Config_005	[128, 64, 32]	0.3	1×10^{-3}	64	0.0227	0.0363	0.9847	51	No
Config_006	[128, 64, 32]	0.2	5×10^{-3}	64	0.0241	0.0382	0.9829	48	No
Config_007	[128, 64, 32]	0.2	1×10^{-3}	128	0.0221	0.0355	0.9858	49	No

$10^{-4}$and more aggressive LR values of 5×10^{-3}. Batch sizes of 32, 64, and 128 samples are evaluated to balance the quality of gradient estimation and computational efficiency. The model validation employs a K-fold cross-validation approach, as described in Eq. (36). In this method, the dataset is divided into K folds, and the model is trained on $(K-1)$ Folds while being tested on the remaining fold. This process is repeated K times so that each fold becomes the test set once. The final cross-validation score is the mean coefficient of determination across all folds. This approach produces a more robust performance estimate than a single train–test split. It uses the entire dataset for both training and validation. It also minimizes the influence of specific data partitions on the performance estimate. As a result, the selected hyperparameters and the model's generalization ability become more reliable and more effective for performance measurement.

4.8.1. Cross-validation

$$K-\text{foldcross}-\text{validationperformance}: CV_{score} = (1/K) \times \Sigma R_k. \quad (28)$$

where R^2_k is the coefficient of determination on fold k, and the sum is from k=1 to K.

The analysis of the 5-fold cross-validation examined results are shown in Table 8. Cross-validation is performed exclusively within the chronologically defined training set and is used for model selection and the final performance is evaluated separately on the held-out test set.

4.8.2. Hyperparameter search space and selected configuration for baseline models

Table 9 summarizes the hyperparameter search spaces and selected configurations for all baseline models discussed in Section 5.2, confirming that each model was systematically tuned using controlled and well-established parameter ranges. The conservative selections reflect a deliberate balance between predictive accuracy, robustness, and computational stability for highly correlated meteorological inputs. Despite this tuning, the baseline models—particularly XGBoost—exhibit inferior accuracy, physics consistency, and uncertainty calibration compared to the proposed physics-constrained framework, indicating that the observed performance gap arises from the absence of embedded physical knowledge rather than insufficient baseline optimization.

4.9. Resource requirements for model deployment

This section examines the computational and memory requirements of the practical implementation of the recommended model architecture. The computational complexity indicates that both the forward and backward passes (N times d times H) are time-bound, where the cost depends on the batch size, feature dimension, and number of hidden units. The uncertainty quantification step, which uses bootstrap sampling, has a complexity of O(B × N × d × H). The memory requirements include storage for model parameters that must be computed at every layer *L* based on Eq. (29). Additional memory is needed for gradient storage, which is twice the parameter memory due to the momentum terms tracked by the Adam optimizer. There is also memory usage for bootstrap storage, computed as B × N × output dimension, to estimate the uncertainty. This type of complexity analysis allows practitioners to approximate the resource demands. It also helps them make well-informed decisions when selecting the model architecture and determining the infrastructure required for deployment.

4.9.1. Computational complexity

Forward Pass Complexity: O (N × d × H)

Backward Pass Complexity: O (N × d × H)

Bootstrap Sampling: O (B × N × d × H) where:

N is the batch size, d is the feature dimension, H is the total hidden units, and B is the bootstrap samples

4.9.2. Memory requirements

Model Parameters: ≈

$$\Sigma(dl \times d_{(l+1)} + d_{(l+1)}) \quad (29)$$

for l=1 to L

Gradient Storage: 2 × model parameters (for Adam optimizer)

Bootstrap Storage: B × N × output dimension

4.10. Training algorithm summary- PCNN training process

Initialize: Network parameters θ, optimizer states
For each epoch: a. **For each batch:**
Forward pass: compute $\widehat{P} = f_\theta(x)$
Compute physics-informed loss: $L_t$otal
Backward pass: compute gradients $\nabla_\theta L_t$otal
Update parameters using Adam optimizer **b. Validation:**
Compute validation loss and metrics
Apply learning rate scheduling if needed
Check early stopping criteria
Output: Trained model parameters θ*

4.11. Uncertainty quantification algorithm- bootstrap uncertainty estimation

1. Input: Trained model $f_{\theta*}$, test data X_{test}
2. For b = 1 to B:
Sample bootstrap dataset D_b* from X_test with replacement
Generate predictions: $P_b = f_\theta*(D_b*)$
Where, B- Number of bootstrap samples
P_b. Prediction from b-th bootstrap model in kW
3. Compute statistics:
Mean prediction: $P_{mean} = (1/B) \times \Sigma P_b$
Standard deviation: $\sigma_p red = \sqrt{(1/(B-1)) \times \sum (\hat{P}_b - \hat{P}_m ean)^2}$
Confidence intervals: $[Q_2.5\%(\hat{P}_b), Q_9 7.5\%(\hat{P}_b)]$
Output: P_{mean}, σ_{pred}, confidence intervals
Bootstrap resampling is applied exclusively to the training (or validation) window using chronologically consistent blocks; the held-out test set is never resampled or used during uncertainty estimation.

This methodology provides a comprehensive framework for implementing PCNN that maintain physical consistency while achieving accurate solar power predictions with quantified uncertainty.

5. Results and discussion

This study presents a comprehensive evaluation of PCNN for PV power generation forecasting through three distinct case studies that progressively demonstrate the model's capabilities and superior performance. Case 1 validates real-world applicability with Chennai meteorological data from Renewables.ninja platform, Case 2 provides a comprehensive comparative analysis against state-of-the-art ML/DL models, including RF, XGBoost, and LSTM networks, and Case 3 establishes foundational PCNN performance using NSRDB benchmark data.

5.1. Case 1: PCNN model validation with the real-time dataset

This section presents the validation of the PCNN model using comprehensive real-time meteorological and solar irradiance values obtained through the Renewables.ninja platform in the Chennai area, Tamil Nadu, India. The validation framework utilizes a total annual dataset of 8760 hourly data (modeled by NASA using the MERRA-2

Table 8
Time-aware K-fold validation results within the training window.

Fold	Training Samples	Validation Samples	MAE	RMSE	R^2	Training Time (min)
Fold 1	35,040	8760	0.0211	0.0344	0.9867	51
Fold 2	35,040	8760	0.0216	0.0351	0.9863	53
Fold 3	35,040	8760	0.0209	0.0345	0.9869	50
Fold 4	35,040	8760	0.0214	0.0349	0.9865	52
Fold 5	35,040	8760	0.0218	0.0353	0.9861	54
Mean	**35,040**	**8760**	**0.0214**	**0.0348**	**0.9865**	**52**
Std Dev	**0**	**0**	**0.0003**	**0.0004**	**0.0003**	**1.6**

Table 9
Hyperparameter search space and selected configuration for baseline models.

Model	Hyperparameter	Search Space	Selected Value	Optimization Method	Remarks
RF	n_estimators	[50, 100, 200]	50	Grid Search + 3-fold CV	Moderate ensemble size
	max_depth	[5, 10, None]	5	Grid Search + 3-fold CV	Controls model complexity
	min_samples_split	[2, 5, 10]	5	Grid Search + 3-fold CV	Regularization
XGBoost	n_estimators	[50, 100, 200]	50	Grid Search + 3-fold CV	Stable boosting
	max_depth	[3, 5, 7]	3	Grid Search + 3-fold CV	Shallow trees to avoid overfitting
	learning_rate	[0.01, 0.05, 0.1]	0.01	Grid Search + 3-fold CV	Conservative learning rate
LSTM	Hidden layers	{[32, 16], [64, 32], [128, 64]}	[64, 32]	Validation-based tuning	Reduced model capacity
	Dropout rate	[0.1, 0.2, 0.3]	0.2 / 0.1	Validation loss minimization	Regularization
	Learning rate	$[1 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}]$	1×10^{-3}	Empirical tuning	Stable convergence

database), ensuring high-quality, satellite-derived verifiability of meteorological inputs that are significant enough in size to cover the globe and support a standard time frame. The validation test is also consistent with Case 1, as the system used is a 10-kW photovoltaic system setup, which involves 40 solar panels, each with 250 W of power. The site of Chennai (13.08°N, 80.27°E) offers some of the most challenging conditions to solar energy models so far because of its tropical wet and dry climate with its own distinct seasonal variations such as the Southwest monsoon (June-September), and Northeast monsoon (October-December), with very specific patterns of condensation and evaporation. These seasonal changes form complicated irradiation patterns with large changes in cloud cover, humidity, and temperature, which provide an extensive challenge to the PCNN model to determine non-linear solar-meteorological interactions thoroughly.

The MAE (right) and training and validation loss (left) after 80 epochs are demonstrated in Fig. 4. The rapid drop in the training curve and validation curve in the initial epochs are good indicators of an effective learning process. It takes approximately 20 epochs before the curves stabilize at a very low value and this is a sign that the model has converged. The training and validation curves nearly align showing good generalization with a small degree of overfit. Validation MAE curve is occasionally oscillating though the oscillations remain within reachable limits. In total, the model has been effective in having low error rates, which shows that it is reliable and that the model is accurate in prediction.

The distribution of the error between the actual and predicted power values is presented in Fig. 5(a). The model also tends to generate non-partisan forecasts, with most residual values usually centered around the zero mark. Higher predicted values of power (more than 4 kW), however, where the residuals do not spread uniformly, but have a few under- and over-estimates, some regular trends are observed. This means that at very high and low prediction ranges, the model works effectively, although its accuracy becomes diminished. The nature of the concentration of points around the zero line confirms that the model accurately describes the underlying relation, with a few variations near certain levels of concentration.

The distribution plot of the residuals (Fig. 5(b)) shows a nearly balanced distribution, with most residuals concentrated around the 0 value. The mean and median residuals (-0.0320) and (-0.0159) are close to zero, implying that there is no observable bias in the model. The weak skewness indicates slight differences in the predictive power and the occurrence of some outliers on either side, which indicate the presence of weak skewness. With most of its errors concentrated around 0, the distribution suggests that the model makes accurate and objective predictions Fig. 5a, Fig. 5b.

Fig. 6 demonstrates the ideal relationship of 1:1 in that the red dashed line compares the values of absolute power, and the value of the model predicted. The coefficient of determination ($R^2 = 0.9983$) in the model is very large, indicating that its predictive power is very high, and

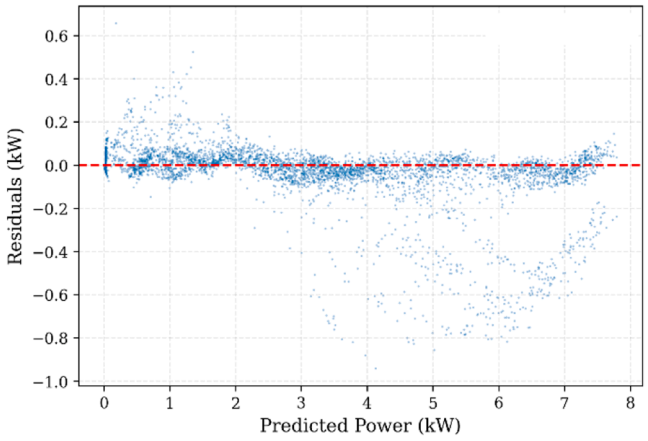


Fig. 5a. Residuals plot of predicted.

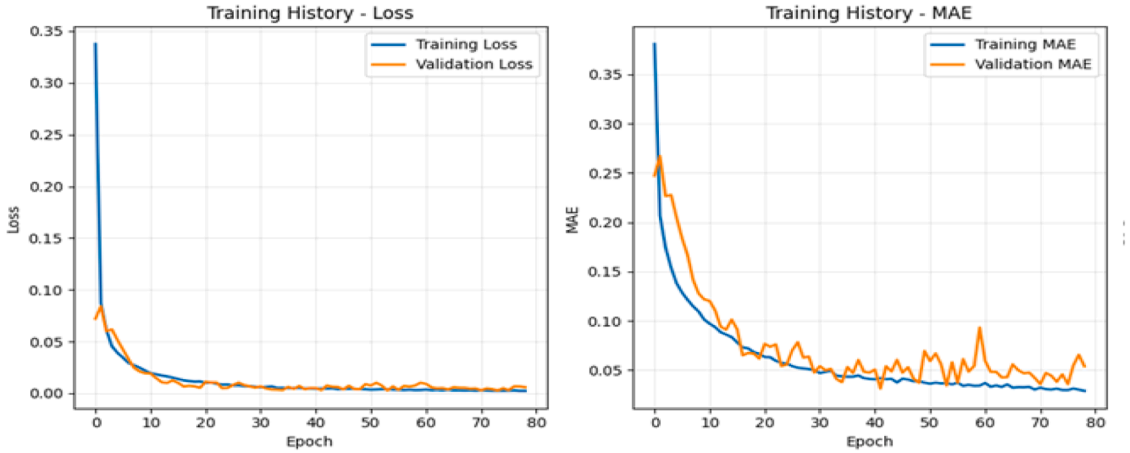


Fig. 4. Training and validation performance: loss and mean absolute error (MAE) across epochs.

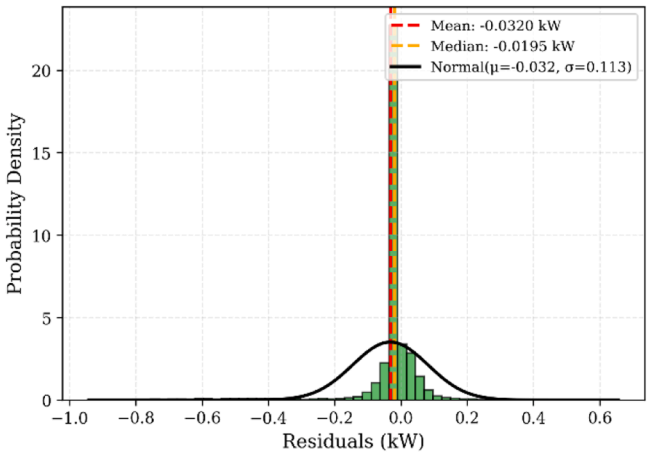


Fig. 5b. Distribution of residuals output for.

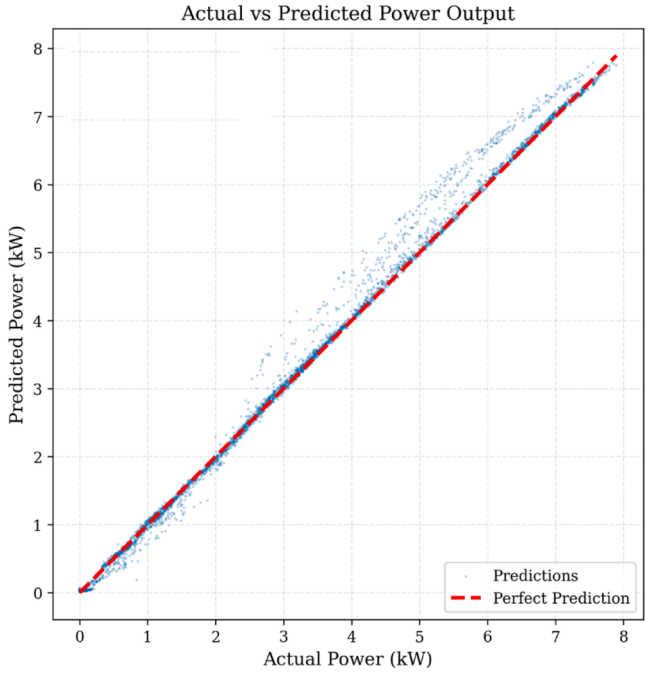


Fig. 6. Actual versus predicted power output with coefficient of determination (R^2).

the points fit the line well. Having minimal variations over the entire prediction range, the near-perfect fit of the model indicates that the model is a true representation of the relationship between the input features and power output.

Fig. 7 shows the performance of a forecasting model on data on power consumption (in kW) over 8760 data points. The visualization provides some critical trends and information regarding the accuracy of the forecast. The real power consumption (in orange) has a high degree of variability and it may vary between almost 0 kW up to peaks of around 8 kW with certain periods of high and low consumption which indicates seasonal or cyclical nature of the energy consumption. The predicted values (blue line) show that the model defines the overall tendencies in the data and helps to give the overall trends of the data and the seasonal tendencies, although with a bit of gouging effect, which is a characteristic of the forecasting models. The 95 % confidence interval (dark blue alternative) is a measure that gives the uncertainty in prediction and it seems to increase at times when there is more volatility in the actual data. It is interesting to note the apparent significant discontinuity in the data where sample index 3000–4000, marked by apparent forecast period in the second half of the time series, with the predictions going past the real-world information which holds them. This model has a decent performance in that it predicts the general amount and season changes of power used but the individual peaks of the pattern are lower than the real observed numbers.

Fig. 8(a) illustrates, how power generation is performed in each month during the 12 months with the actual data (blue line with circles) and the projected data (orange line with triangles). This data indicates a definite U-shaped seasonal pattern in power generation whereby the highest rate of power production is registered during the winter season (January-February) at approximately 2.2 kW and which then sharply drops in early spring and lowest production is observed in summer (May-July) at about 1.1 kW. Generation of power then monotonically decreases between August till December and then drops again to high winter levels. The forecasting model has high accuracy in dispersing the current seasonal trend with the values forecasted strongly following the real- measurements in all months. The model manages to determine the timing and the magnitude of the peak and the trough time periods with not many deviations. This seasonal effect is a typical behaviour of heating dominated energy models or renewable sources of energy such as solar power sources in regions of specific geographical locations where the time of winter produces higher generation since the heat demand is greater or the weather is favourable. The high correlation of actual versus predicted values indicates that the underlying seasonal drivers have been learned well by the model and can confidently predict the monthly generation pattern to build up a forecast on either a long-range basis (planning) or a short-term basis (operations).

The pattern of the daily power generation over 24 hours as illustrated in Fig. 8(b) depicts the average power output of a day. The general distribution of the data is a bell-shaped curve with power generation at zero at night (0–5 and 18–24), followed by slow increase of the same to

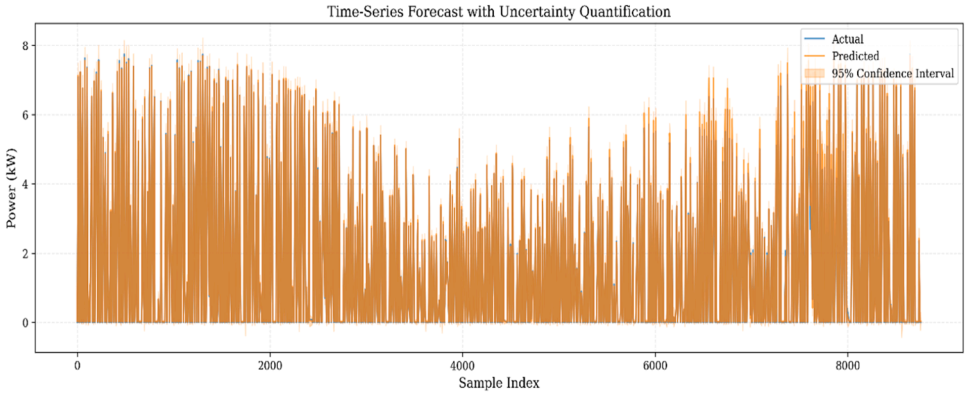


Fig. 7. Time series forecasting analysis with confidence intervals.

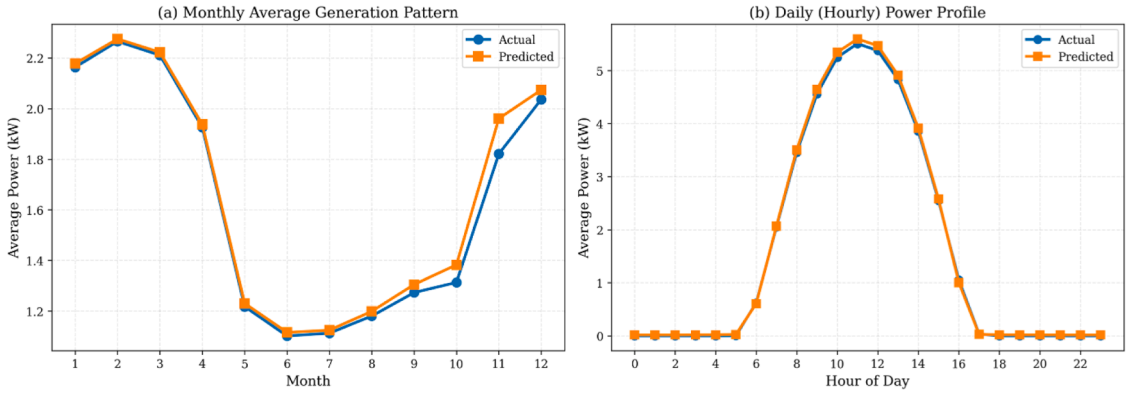


Fig. 8. (a) Monthly average power generation pattern: actual vs predicted values, 8(b) Daily power generation profile: hourly pattern analysis (actual vs predicted).

5.5 kW around mid-day (hour 11–12), and then reaching a steady decrease during the afternoon and evening hours. This trend is very lucrative to solar power generation based on photovoltaic technology where the rate of energy production is directly proportional to sunshine and the amount of sunlight irradiating the earth during the day. The forecasting model is significantly accurate in the prediction of this daily generation profile and the values (orange line) of the forecasting model exactly overlap the actual measurements (blue line) at all hours. The model is useful in its ability to capture the timing and intensity of the generation curve including the smooth transitions of the sunrise or sunset periods. The fact that the actual and predicted figure are very close shows that the solar resource trends are acquired by the model and can be trusted to make accurate projections of hourly generation to day-ahead planning, grid integration, and energy management uses. Another aspect that alludes to the consistency in weather conditions and better placement of solar panels to maximize sun power is the symmetrical nature of the curve towards the middle of the day.

The scatter plot in Fig. 9 shows how predicted power relates to the prediction uncertainty of the same (both in kW units). This trend indicates that at medium values of power, especially at the range of about 4 kw, there is low uncertainty in prediction and where the trend is likely to rise towards the low and high limits of the intended power. This implies that when the predictive model is predicting at the middle of the power values, the prediction has more confidence (the model is less uncertain). Simultaneously, towards the edges of the prediction space, its confidence declines (uncertainty increases). The given trend is prevalent in most real-world regression models whereby lack of available data or an augmentation of complexity in the far ends of the variable results in increased uncertainties in predictions. This observation shows the significance of uncertainty consideration, in particular, in performing operational decisions at the tip-ends of the forecasted power range.

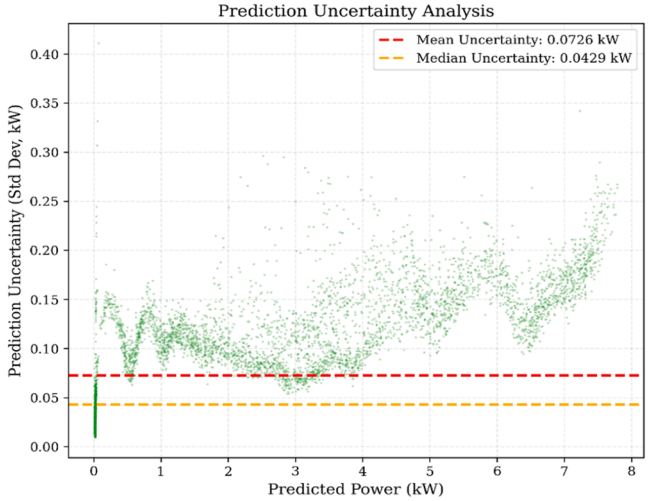


Fig. 9. Prediction Uncertainty as a Function of Predicted Power.

Fig. 10 shows the seasonal and monsoon performance analysis. The predictive power is consistently strong, despite varying weather conditions, with R^2 values ranging from 0.9859 to 0.9997, which strongly indicates that the model performs well across different weather conditions. The use of summer conditions (Fig. 10 (b)) yields the highest prediction accuracy ($R^2 = 0.9997$), with close performance from winter (Fig. 10 (a), $R^2 = 0.9992$), indicating that the model performs well in extreme conditions of the year, characterized by highly seasonal and relatively predictable weather patterns. Fig. 10 (c) indicates good predictive behaviour of the Southwest Monsoon, with a high predictive score ($R^2 = 0.9990$) and a close cluster of full-bellied predictive data, indicating good performance even in extreme conditions of heavy rains. The Northeast Monsoon (Fig. 10 (d)) exhibits somewhat lesser yet still satisfactory performance (R2 = 0.9809), with slightly higher scatter between the predicted and actual power values. This may be due to the more variable and unpredictable weather patterns during this period, especially during the monsoon season. All in all, the scatter plots indicate that the model remains highly faithful to all seasonal variations, with the predicted power values being very close to the actual power production throughout the entire range of operational helpfulness of the model, which further justifies the strength and dependability of the model in power predictions that may be wide-ranging and year-round.

5.2. Case 2: multi-dimensional performance analysis: PCNN vs. traditional ML/DL models for solar power forecasting

The section entails an elaborate discussion of several ML and DL models in predicting solar power generation with a special focus on PCNN as a unique model. The paper is a comparative study of PCNN with other models to determine the model's accuracy, robustness, physics consistency, computational efficiency, and adaptability to various situations. The analysis utilizes actual solar power measurements from Chennai, India, to provide practical information for optimizing the renewable energy forecasting system. This analysis was undertaken in response to the growing demand for reliable solar energy solutions as the world transitions toward sustainable power sources. The accurate prediction of solar power is essential for grid stability, energy market planning, and resource allocation, but traditional models may struggle to cope with the complex physics-based dynamics of PV systems. The neuron network architecture of PCNN, which incorporates physical principles, is a promising alternative that enhances the precision of predictions and reliance on the underlying physical laws. The initial goal of the study is to meticulously compare PCNN with traditional ML and DL models, their advantages, and flaws, so that real-world applications can adopt the former. Furthermore, the analysis addresses

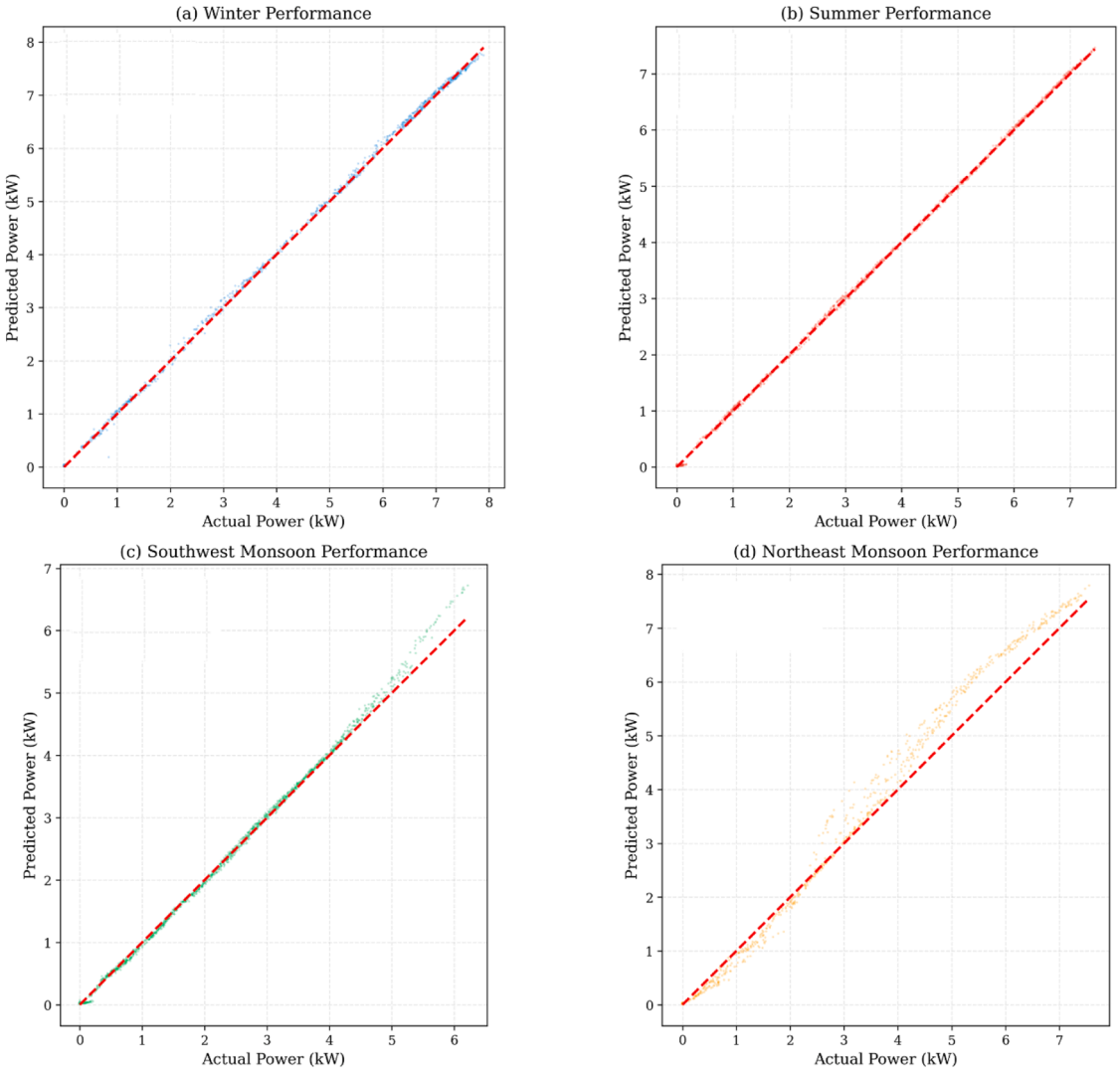


Fig. 10. Seasonal performance evaluation of the PCNN Model: (a) Winter, (b) Summer, (c) Southwest Monsoon, and (d) Northeast Monsoon.

major pitfalls, including data scarcity, out-of-distribution robustness, and computational feasibility, providing a comprehensive assessment to inform decision-making in the renewable energy industry.

According to the results in Table 10, the PCNN model achieved the best performance, with minimal error values (RMSE = 0.0448, MAE = 0.0270) and the highest coefficient of determination (R^2 = 0.9986), indicating very accurate predictions. On the contrary, the RF model demonstrated moderate accuracy (R^2 = 0.9895), while the LSTM showed excellent performance (R^2 = 0.9976), albeit with fewer errors. The XGB model ranks lowest (R^2 = 0.9355), and its errors, as measured by MAE, are significantly large, making it the least effective model to use. These findings indicate the effectiveness of the PCNN in modeling system dynamics and minimizing prediction errors.

5.2.1. *Uncertainty quantification analysis*

The uncertainty estimates produced by the proposed framework provide actionable information beyond point forecasts. Narrow prediction intervals during stable irradiance conditions indicate high forecast confidence and can support reduced reserve margins and more aggressive energy dispatch decisions. Conversely, wider intervals observed during low-irradiance periods, rapid weather transitions, or extreme operating conditions signal increased forecast risk and can be used to trigger conservative operational strategies, such as higher spinning reserve allocation or flexible load scheduling. The separation of epistemic and aleatoric uncertainty further enables operators to distinguish between model-related uncertainty, which may be reduced through additional data or retraining, and inherent environmental variability, which is largely irreducible. Overall, these probabilistic outputs enhance situational awareness and support risk-informed planning and operation of PV-integrated power systems.

The results of the uncertainty quantification presented in Table 11 illustrate the superiority of the PCNN in terms of the trade-off between

Table 10
Prediction accuracy comparison.

Model	R^2	RMSE	MAE
PCNN	0.9986	0.0448	0.0270
LSTM	0.9976	0.1125	0.0591
RF	0.9895	0.1491	0.0851
XGB	0.9355	0.2333	1.3929

Table 11
Uncertainty quantification analysis.

Model	PICP	MPIW	CWC	Reliability	CRPS
PCNN	0.9472	0.3127	0.1168	0.0028	0.0137
LSTM	0.8765	0.4281	0.4693	0.0735	0.0284
RF	0.9138	0.2536	0.2784	0.0362	0.0341
XGB	0.908	0.1062	0.5104	0.037	0.0678

coverage of prediction intervals, sharpness, and reliability. The PCNN recorded the best PICP (0.9472), which is very close to the target of 95 % confidence level, meaning that the intervals used are effective in taking accurate outputs, and the widths are also narrow (MPIW = 0.3127). The value of its CWC (0.1168) and reliability (0.0028) are the lowest among the rest, indicating that PCNN provides a calibrated interval with minimal variance and nominal coverage. Notably, PCNN achieved the highest CRPS (0.0137), indicating maximum good performance in probabilistic accuracy, as sharpness and calibration are combined into a single value. On the contrary, LSTM did not perform well with lower coverage (PICP = 0.8765) and the broader intervals (MPIW = 0.4281), resulting in a significantly greater CRPS (0.0284). The tree-based models (RF and XGB) produced smaller intervals; however, their PICP (0.9138 and 0.908) and CRPS (0.0341 and 0.0678) were lower, suggesting a lack of uncertainty estimation and unreliability in calibration.

5.2.2. Physics consistency analysis

Table 12 shows that the PCNN model has the strongest correlation (Phys_Corr = 0.9989) and the lowest violation rate (0.10 %, just nine negative violations, and no high violations), according to the physics consistency study. Although the LSTM also exhibits a strong physical correlation (0.9981), it frequently violates physical limitations, as evidenced by its significantly higher violation rate (50.45 % with 4419 negative violations). Although the RF model records a minor violation rate (0.22 %, 30 negative violations), it obtains a good correlation (0.9790) and relatively low error (Phys_RMSE = 2.7842). With the highest violation rate (0.50 %, 43 negative violations, including some serious violations) and the lowest physical correlation (0.9244), the XGB model exhibits the poorest performance. Overall, these results confirm that PCNN ensures the strongest physics-consistent predictions, while LSTM and XGB suffer from significant constraint violations.

Table 13 shows Out-of-Distribution Robustness Comparison of PCNN, RF, LSTM, and XGB Models Under Various Environmental Conditions. The results show that the PCNN model exhibits near-perfect correlations at high temperatures (0.9983), low temperatures (0.9985), nighttime hours (0.9844), and peak summer (0.9996), according to the out-of-distribution robustness analysis. However, it performs worse under high irradiation (0.3630). The RF model is a trustworthy baseline since it performs similarly well under most circumstances and maintains strong robustness in all situations, including heavy irradiation (0.8339). The LSTM model performs well in temperature and seasonal extremes, but it is sensitive to challenging conditions, showing instability under high irradiance (5.21093) and low resilience at night (0.1670). The XGB model performs consistently worse than PCNN and RF, especially at night (0.9108) and during the hottest summer (0.9406), although it offers moderate robustness in all conditions. Overall, PCNN and RF stand out as the most robust models, with PCNN excelling in extreme temperatures and during seasonal peaks. At the same time, RF provides stronger stability under high irradiation and night conditions.

5.2.3. Robustness analysis

According to Table 14, the data scarcity robustness analysis shows that the PCNN model performs well at all data reduction levels. Its accuracy values range from 0.988 at 30 % data availability to 0.996 at 90 %, demonstrating exceptional resilience even in the face of restricted data.

Table 12
Physics consistency analysis.

Model	Phys_Corr	Phys_RMSE	Viol_Rate	Neg_Viol	High_Viol
PCNN	0.9989	2.8037	0.10	9	0
LSTM	0.9981	2.8494	50.45	4419	0
RF	0.9790	2.7842	0.22	30	0
XGB	0.9244	2.6154	0.50	43	0

Although it performs somewhat worse than PCNN, the LSTM model also exhibits excellent robustness, increasing from 0.962 (30 %) to 0.985 (90 %). At higher scarcity levels, the RF model does not achieve the same reliability as PCNN, but it maintains a consistent level of stability and achieves good accuracy (0.978–0.991). The XGB model is most impacted by data scarcity, exhibiting reduced flexibility in such conditions. It starts at 0.922 (30 %) and only reaches 0.962 (90 %). Overall, the results confirm that PCNN is the most robust model under data scarcity, followed by LSTM and RF, while XGB shows the weakest resilience.

5.2.4. Computational efficiency analysis

Table 15 presents the Computational Efficiency of different models. The PCNN model requires the longest training time (343.61 s) and the highest inference time (0.3928 s) compared to other models, as determined by the computational efficiency analysis. This is primarily due to its physics-informed architecture, which combines data-driven learning with physical constraints, thereby significantly increasing computational complexity. With a training duration of 107.42 s, the LSTM also requires a significant number of resources, which is indicative of the sequential nature of its recurrent structure. Traditional machine learning models, on the other hand, such as RF (2.82 s) and XGB (0.35 s), are computationally lightweight since they train considerably faster and require less inference time. However, PCNN's greater prediction accuracy, physical consistency, and robustness under difficult conditions, qualities that the simpler models cannot match, justify the additional expense of training it.

5.2.5. Convergence and stability analysis

Table 16 illustrates the Convergence and Stability Analysis of Neural Network Models. The results show that the PCNN model exhibits very low overfitting (0.5226) and achieves superior convergence, with a larger loss reduction (0.9736) and strong stability (0.000017) over 50 epochs, as indicated by the convergence and stability analysis. However, even though the LSTM model converges more quickly within 26 epochs, it exhibits a much greater overfitting rate (10.3009) and a lower convergence efficiency (0.9251), indicating a lesser capacity for generalization than the PCNN. This explains PCNN's accuracy and physics consistency resilience. Since the RF and XGB models are tree-based, non-neural network techniques that do not learn via epoch-based convergence or iterative optimization, they are excluded from our research. As a result, these models cannot directly utilize convergence criteria such as overfitting, stability, and loss reduction.

The PCNN model demonstrates superior performance compared to the baseline ML/DL models in all types of analysis and is therefore selected for the subsequent analysis in Case 3.

5.3. Case 3: PCNN training convergence, model validation, and seasonal performance analysis

Fig. 11 illustrates the training curve of the learning process of a PCNN in 30 epochs. The model exhibits a high convergence rate, with the training loss (red line) initially at approximately 0.055 and decreasing significantly in the first five epochs before reaching its current level of 0.002–0.003. The loss in validation (yellow line) is another similar line in that it starts at low values of around 0.005 and soon reaches similar low-fond values. The sign of this behaviour is that the PCNN was able to acquire the underlying physics constraints and data patterns extremely efficiently. Low training and validation loss coincide rates over the training process indicate a good generalization capacity and no chance of overfitting. The value of low losses, first achieved by epoch five and maintained thereafter, demonstrates that the model has found an optimal solution relatively quickly during training and has remained stable, which is typical of well-constructed PCNNs. These models can efficiently apply physical laws during the learning process.

Fig. 12 presents two complementary plots that evaluate the performance of the PV power prediction model. Fig. 12 (a) shows the

Table 13
Out-of-distribution robustness comparison.

Model	High_Temp	Low_Temp	High_Irradiation	Night_Hours	Peak_Summer
PCNN	0.9983	0.9985	0.3630	0.9844	0.9996
LSTM	0.9550	0.9858	5.2093	0.1670	0.9900
RF	0.9972	0.9983	0.8339	0.9967	0.9995
XGB	0.9710	0.9814	0.9421	0.9108	0.9406

Table 14
Data scarcity robustness.

Model	30 %	50 %	70 %	90 %
PCNN	0.988	0.995	0.995	0.996
LSTM	0.962	0.978	0.986	0.985
RF	0.978	0.9812	0.990	0.991
XGB	0.922	0.962	0.96	0.962

Table 15
Computational efficiency of different models.

Model	Train_Time	Inference	Memory
PCNN	343.61	0.392780	TensorFlow
LSTM	107.42	0.217382	TensorFlow
RF	2.82	0.000586	Scikit-learn
XGB	0.35	0.022219	XGBoost

Table 16
Convergence and stability analysis of neural network models.

Model	Loss Reduction	Stability	Overfitting	Epochs
PCNN	0.9736	0.000017	0.5226	50
LSTM	0.9251	0.000008	10.3009	26

forecasted values against the actual PV power values. The points closely follow the ideal diagonal line (red dashed line), which indicates high prediction accuracy. The distribution of points is narrow and concentrated around the optimal prediction line. This suggests that the model has effectively captured the internal relationships between the input attributes and PV power output, with minimal systematic error. Fig. 12 (b) shows the residuals, providing a deeper understanding of the model's behaviour. The residuals are well distributed around zero (red dashed horizontal line), indicating that the predictions are not biased. However, some heteroscedasticity appears at higher predicted power values, particularly in the (100–175) W range. The magnitudes of the residuals generally fall within ±40 W, which is reasonable for PV power forecasting.

Overall, these two plots together show that the model is robust across the full operating range. The residual analysis confirms that the forecasting errors are scattered without any significant systematic pattern, indicating the model's strong effectiveness.

Fig. 13 illustrates the seasonal patterns of power production from PV stations from 2016 to 2020, which are presumably estimated or simulated using a PCNN model. The information shows that seasonal variations are consistent across all five years, with spring generating the best power output (in the range of 150–155 kWh yearly) and summer close behind (in the range of 120–125 kWh yearly). Both winter and fall have lower but comparatively stable generation, with winter generation at about 115–125 kWh and the lowest generation during fall years at about 100–120 kWh. The fact that the PCNN model incorporates physical laws governing solar irradiance, weather, and the behaviour of the PV system would have allowed the model to capture these seasonal variations. It is interesting to note that the generation is slightly higher in 2019 in most seasons, especially in spring, which may indicate either favourable weather conditions or more efficient system operation in that year. The overall similar seasonal trend of the two years confirms the possibility of comprehending and predicting the underlying physics-based correlations between the seasonal variables (solar angle, daylight, temperature, weather conditions) and PV power generation using the PCNN model. This seasonal analysis demonstrates the model's ability to provide trustworthy data on energy planning and system optimization across various seasons of the year.

Fig. 14 shows the probability distribution of the output power of the photovoltaic system as input to an estimating or analyzing model using a PCNN, with the panel capacity rating of 250 W indicated by a red dashed vertical line. The power generating characteristics are highly skewed, with the most concentration at very low power outputs (almost at 0 W), meaning that the PV system also works at low or no generation conditions very often. This trend is characteristic of real-life installations of the solar system due to the hours of darkness, cloudiness, and the low light hours that comprise most of the operating schedule. The PCNN model accurately represents the reality of physics-based models, which indicate that solar panels spend too much time in low-irradiance conditions and that the power output of a solar panel declines rapidly with increasing frequency towards high values. The distribution is steeply

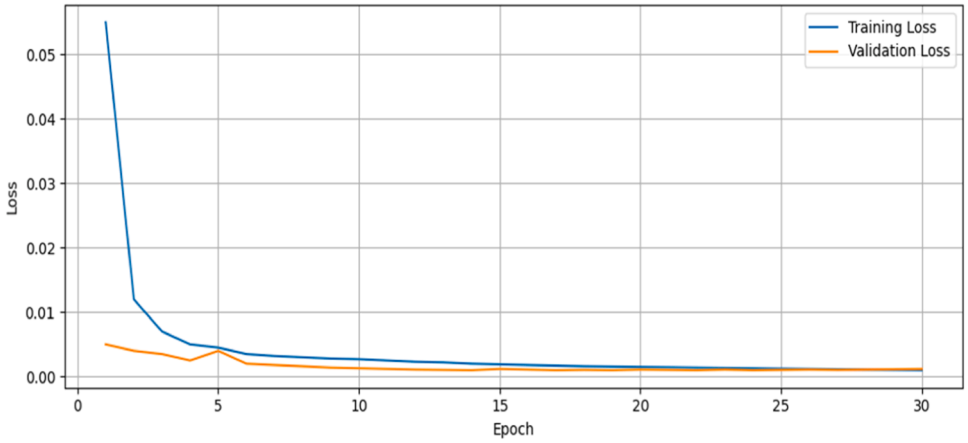


Fig. 11. PCNN training and validation loss convergence.

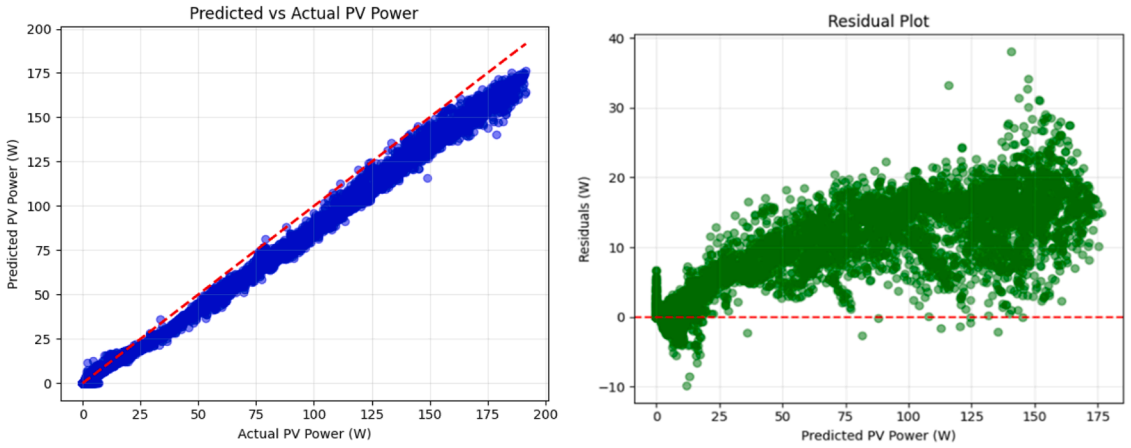


Fig. 12. Predicted vs actual PV power performance and residual analysis for PV power prediction.

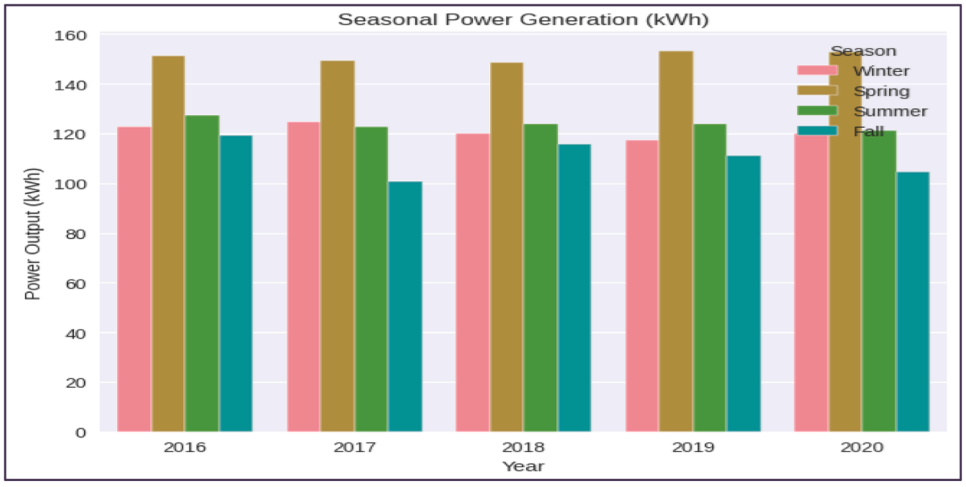


Fig. 13. Seasonal solar energy generation patterns using PCNN methodology.

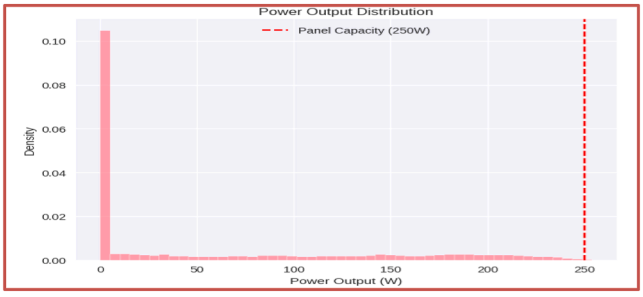


Fig. 14. PV power output probability distribution analysis.

decreasing at the maximum density of 0 W, with extremely spaced cases of generating power that is near the maximum capacity of the panel, i.e., 250 W. This is a measure of the physical limitations placed on the generation of solar power, in which optimum conditions (maximum sun time, unobstructed skies, and the panels facing the sun) only constitute a tiny percentage of the overall operating time. The capability of the PCNN to recreate this realistic pattern of power distributions attests to its ability to incorporate not only meteorological physics but also those of PV systems, which can be utilized in energy system planning, storage sizing, and performance optimization plans.

Fig. 15 shows a boxplot that depicts the entire winter patterns of monthly PV power generation, analyzed using a PCNN model. It reveals the obvious seasonal patterns in all months of the year. The highest level of generation is observed during the spring months (March-May), with median figures situated at 50–53 kWh, and distributions that are not very broad, suggesting that it is controlled by constant high-performance periods. The generation in the summer (June-August) is moderate with a median of 41–43 kWh and a high variability, which

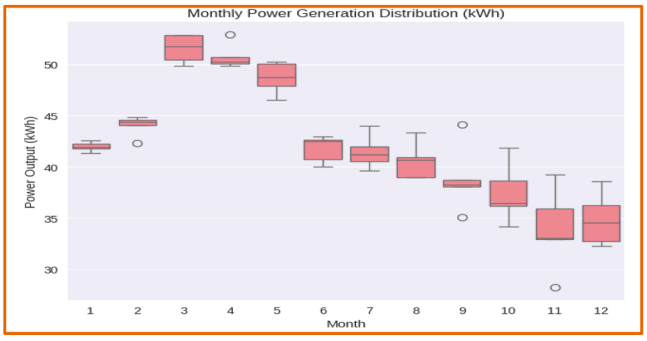


Fig. 15. PCNN-predicted monthly PV energy distribution with outlier detection.

may be caused by the variability of weather patterns in hot seasons. The minimum generation occurs in winter (November-February), with a median of 35–45 kWh, accompanied by some significant outliers, which highlights the effects of lower daylight hours and poor weather conditions. The PCNN model is beneficial for capturing physics-based seasonal solar irradiance patterns, particularly in spring, when moderate temperatures and increasing daylight make it optimal. Winter exhibits the anticipated decrease in solar energy. There are a few months that show outliers, especially in January, April, June, September, and December, as these months occasionally experience extreme weather events or special performance days, which the model can account for, and this is reflected in its forecast.

5.3.1. Comparative performance analysis: PCNN vs. benchmarked ML models

The given analysis compares the proposed PCNN model to the existing benchmarked algorithms suggested by A. Patel and O. V. Gnana Swathika (2024) for predicting annual data. The same dataset was used to train and evaluate all models, ensuring they are fair and without bias compared. The performance measures are various metrics that assess the accuracy and reliability of the model in predictive modelling applications.

The comparative analysis, as shown in Table 17, Figs. 16 and 17, indicates that the proposed PCNN model exhibits a high level of performance compared to the traditional benchmarked models in the same training and testing datasets. The PCNN model has the lowest Mean Absolute Error (MAE) of 0.0213, which is significantly lower than that of the SVR (0.0552), Extra Tree (0.0310), Gaussian Process Regression, and Neural Network (0.0482) models. Likewise, PCNN has a Root Mean Square Error (RMSE) of 0.0347, which is significantly smaller compared to all the benchmarked alternatives, and the closest performance is recorded by the Neural Network at 0.0592. The coefficient of determination (R^2) further demonstrates the superiority of PCNN, with a value of 0.9865, indicating that the model explains approximately 98.65 % of the variance in the data. This is significantly higher in performance than that of benchmarked models, and Extra Tree has the highest R^2 of 0.9740. These results collectively demonstrate that the physics-informed approach effectively leverages domain knowledge to enhance predictive accuracy and model reliability, achieving substantial improvements over conventional machine earning methods when applied to the same underlying dataset.

5.3.2. Limitations and scope of applicability

The proposed PCNN demonstrates strong performance and physical consistency across the evaluated datasets; several limitations should be acknowledged. First, the primary analyses are based on meteorological data derived from NSRDB and Renewables.ninja for a limited set of geographic and climatic conditions. Although these datasets are widely used and representative, the results may not fully capture site-specific effects such as complex shading, soiling, or rapid microclimatic variations observed in other locations.

Second, the physics-informed component of the proposed framework adopts a constraint-guided formulation rather than explicitly solving governing differential equations of photovoltaic systems. While this approach effectively enforces operational limits and improves physical plausibility, it does not model detailed semiconductor-level or electro-thermal dynamics, which may be relevant for component-level diagnostics or fault analysis.

Third, although robustness across multiple temporal scales and operating conditions is demonstrated, the generalizability of the framework to different PV technologies, system sizes, and climates has not been exhaustively validated. Further multi-site and multi-technology studies are required to establish broader applicability. Finally, the incorporation of physics constraints and uncertainty quantification increases model complexity and computational overhead compared to purely data-driven models. While the proposed framework remains suitable for offline forecasting and planning applications, additional optimization may be required for large-scale real-time deployment.

Despite these limitations, the proposed framework provides a structured and extensible foundation for physics-constrained and uncertainty-aware PV power forecasting, and future work will focus on expanding validation across diverse sites, integrating richer physical models, and improving computational efficiency.

Table 17
Performance evaluation of PCNN model against benchmarked machine learning algorithms: a comparative analysis of error metrics and predictive accuracy.

Reference	Model	MAE	MSE	RMSE	R^2
A. Patel and O. V. Gnana Swathika., (2024) [51]	Support Vector Machine	0.0552	-	0.0707	0.9370
	Extra Tree	0.0310	-	0.0454	0.9740
	Gaussian Process Regression	0.0482	-	0.0694	0.9393
	Neural Network	0.0408	-	0.0592	0.9557
Proposed	**PCNN**	**0.0213**	**0.0903**	**0.0347**	**0.9865**

5.3.3. Failure modes and challenging forecasting scenarios

Although the proposed framework demonstrates robust performance under the evaluated conditions, certain challenging scenarios may lead to reduced accuracy. These include rapid cloud transients and abrupt irradiance fluctuations that are difficult to capture using hourly meteorological inputs, as well as rare or extreme atmospheric conditions that are underrepresented in the training data. In addition, model performance may degrade when applied to PV systems with significantly different configurations, such as alternative module technologies, pronounced shading, or unmodeled degradation effects. While physics-based constraints mitigate physically implausible predictions, they do not fully capture all site-specific nonlinearities, highlighting the need for further validation under diverse operating conditions.

6. Conclusion

The widespread deployment of PV systems presents significant challenges for grid stability due to the inherent variability of solar resources. Accurate forecasting with reliable uncertainty estimation is essential, yet traditional ML and DL methods often overlook the underlying physics of PV systems and lack robust probabilistic outputs. This study proposes a PCNN framework that embeds photovoltaic physics, thermodynamic principles, and energy conservation laws directly into the neural architecture while incorporating Bayesian and bootstrap-based uncertainty quantification to capture both epistemic and aleatoric uncertainties. The framework is validated through three case studies: (i) real-time performance evaluation using meteorological inputs from the Renewables.ninja platform, (ii) benchmarking against advanced ML/DL models such as RF, XGBoost, and LSTM, and (iii) analysis using synthetic climatic data from the NSRDB with uncertainty calibration. Across all cases, the proposed PCNN demonstrates superior predictive accuracy, achieving an MAE of 0.0213, an RMSE of 0.0347, and an R^2 of 0.9865 on the NSRDB data and an R^2 of 0.9983 on real-time data from Chennai, outperforming baseline models, including LSTM, RF, XGB, SVR, and neural networks. The uncertainty quantification analysis demonstrates high reliability, with a PICP of 94.72 %, MPIW of 0.3127, and a low CRPS of 0.0137, indicating well-calibrated and sharp intervals. Physics-consistency evaluation further confirms the model's robustness, with the PCNN exhibiting the lowest violation rate (0.10 %) and the strongest adherence to physical constraints compared to benchmarked models. Seasonal, monsoon, and stress-condition analyses highlight the model's resilience across extreme weather, high-irradiance events, night-time conditions, and out-of-distribution scenarios. By

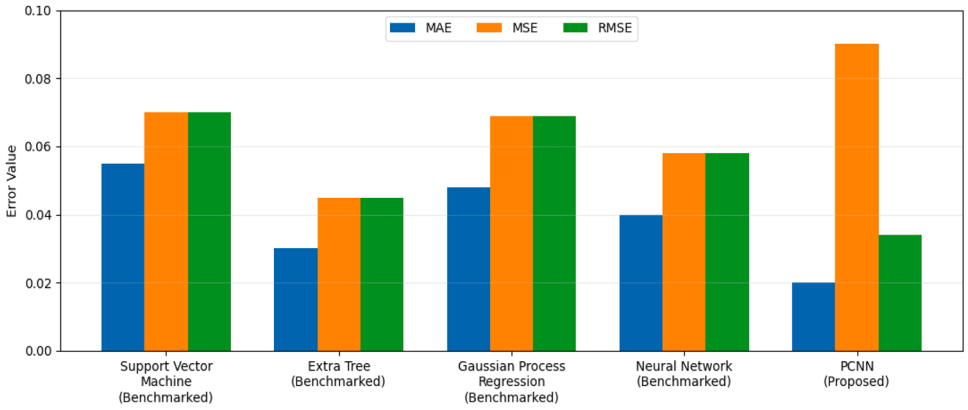


Fig. 16. Comparative analysis of error metrics (MAE, MSE, RMSE) across machine learning models.

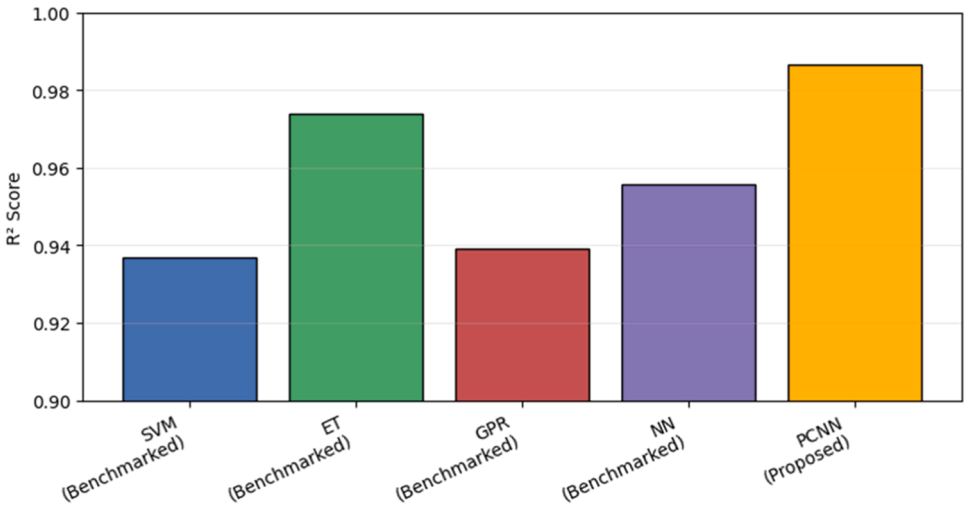


Fig. 17. Coefficient of determination (R^2) comparison: PCNN model vs. benchmarked algorithms.

integrating physical interpretability, high predictive accuracy, and quantified uncertainty, the proposed PV- PCNN framework supports more reliable decision-making for grid integration, energy trading, reserve planning, and smart grid operations.

Looking ahead, several promising directions emerge from this work. First, extending the framework to hybrid renewable systems that couple solar with wind or storage could enhance its applicability to integrated energy systems. Second, incorporating spatiotemporal satellite imagery and remote sensing data within the PCNN architecture may improve spatial generalization and forecasting accuracy across larger geographic regions. Third, the exploration of adaptive or self-tuning physics constraints could enable the model to balance data fidelity and physical consistency under evolving conditions dynamically. Finally, deploying the PCNN in real-time energy management platforms with edge or cloud-based integration would bridge the gap between research and large-scale industrial adoption. Collectively, these future directions highlight the potential of physics-informed machine learning to become a cornerstone methodology for next-generation renewable energy forecasting and sustainable grid operations.

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Data availability

Researchers can access the identical dataset through the NSRDB Data Viewer platform (https://nsrdb.nrel.gov/data-viewer) for verification, replication, or extended analysis purposes.

Renewables.ninja PV Data (MEERA-2 Dataset): PV resource data used in this study was sourced from Renewables.ninja, an open-access platform for renewable energy modeling data. The dataset is accessible at: https://www.renewables.ninja

Ethics declaration

Not applicable.

CRediT authorship contribution statement

Rajaperumal T. A: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Christopher Columbus Chinnappan:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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